

Lesson Plan

Subject Name:- data warehousing data mining	Class:- B.Tech CSE	Subject Code: CIE-425T
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32	Data Warehouse and Business Strategy.	1	paulraj punniah		
33	UNIT-3 Data Mining Basics: What is Data Mining, Data Mining Defined, The knowledge discovery process (KDD Process),	1	paulraj punniah		
34	Data Mining Applications- The Business Context of Data Mining	1	paulraj punniah		
35	Data Mining for Process Improvement, Data Mining as a Research Tool,	1	paulraj punniah		
36	Data Mining for Marketing, Benefits of data mining,	1	paulraj punniah		
37	Data Mining Techniques: Classification and Prediction: Issues Regarding	1	paulraj punniah		
38	Classification and Prediction, Classification by Decision Tree Induction, KNN Algorithm.	1	paulraj punniah		
39	UNIT-4 Cluster detection, K- means Algorithm	1	paulraj punniah		
40	Outlier Analysis, memory-based reasoning	1	paulraj punniah		
41	Link analysis, Mining Association Rules in Large Databases	1	paulraj punniah		
42	Association Rule Mining	1	paulraj punniah		
43	Genetic algorithms	1	paulraj punniah		
44	Neural networks. Data mining tools.	1	paulraj punniah		

Textbook(s):

Data Warehouse & Data Mining [Nitasha Rathore]

1. Paul Raj Poonia, "Fundamentals of Data Warehousing", John Wiley & Sons, 2004
2. Kamber and Han, "Data Mining Concepts and Techniques", Hart Court India P. Ltd. Elsevier, 2nd Ed, 2001

References:

1. W. H. Inmon, "Building the operational data store", 2nd Ed., John Wiley, 1999.
2. Pang- Ning Tan, Michael Steinbach, Viach, Vipin Kumar, Introduction to Data Mining, Pearson
3. Shmueli, "Data Mining for Business Intelligence : Concepts, Techniques and Applications in Microsoft Excel with XLMiner", Wiley Publications

1. Introduction to Data Warehousing and data mining

Data warehousing refers to a typical procedure of compiling and organising data into a common database. On the other hand, data mining basically refers to the process of extracting useful data from various databases.

Please note that the data mining procedure entirely depends on the data that is compiled within the data warehousing phase. This helps in recognising all the meaningful patterns of the data. In fact, data warehousing is formulated to provide aid to management systems.

What is Data Mining?

In the process of data mining, data is carefully extracted and analysed to fetch nothing other than useful information. Here, all the hidden patterns are researched from the dataset, thereby predicting future behaviour. Besides, it can seamlessly indicate and discover unique relationships through the data.

On the other hand, data mining makes the best use of artificial intelligence, statistics, machine learning systems, databases, etc. It is used for figuring out the hidden patterns within the data. Additionally, it also supports all business-related queries that take loads of time to resolve.

What is Data Warehousing?

A data warehouse is nothing but an electronic storage that stores gigantic amounts of business information. It is exquisitely designed for both query and analysis rather than processing transactions.

Data warehousing is a unique technique that helps collect and manage data from various sources. Furthermore, these data provide meaningful and valuable business insights. It is basically a blend of components and technologies that enables strategic usage of data.

In a nutshell, data warehousing is a process of transforming data into information, thereby making it widely available for analysis. Now, let us discuss the important differences between data mining and data warehousing in detail.

Features of Data Warehouses

Here is a comprehensive listing of the most common features of data warehouses:

1. Unified

Numerous heterogeneous sources are compiled together to formulate a data warehouse like social databases, level documents, etc.

2. Subject Oriented

It offers the most important data regarding particular subjects like products, suppliers, customers, promotions, etc. It handles data analysis and modelling while assisting organisations in making data-driven decisions

3. Nonvolatile

The previous data remains intact whenever new data is added to the data warehouse. Since the operational database and data warehouse are kept separately, changes within the operational database aren't shown.

4. Time-Variant

Data collected within the data warehouse is seamlessly identified within a particular period.

Features of Data Mining

Data Warehouse & Data Mining [Nitasha Rathore]

Some of the unique features of data mining are:

- It is capable of predicting future results
- It can efficiently handle large datasets and databases
- It can seamlessly utilise the automated discovery of patterns
- It has the potential to create actionable insights, etc.

Applications of Data Mining

The applications of data mining include:

- Research
- Education Sector
- Transportation
- Market Basket Analysis
- Business Transactions
- Intrusion Detection
- Scientific Analysis
- Finance and Banking Sector
- Insurance and Healthcare

Applications of Data Warehouses:

The applications of data warehouses include:

- Banking Services
- Consumer Goods
- Manufacturing
- Financial Services
- Retail Sectors

Benefits of Data Mining

Here, we have listed a few potential benefits of data mining:

- Analysing trends within the existing marketplace.
- Detecting typical frauds in phone calls, insurance claims, debit or credit purchases, etc.
- It can make easy predictions within the market before making business decisions.

Benefits of Data Warehousing

The most compelling advantages of data warehousing are:

- Improved performance and productivity
- Cost-effective
- Consistent and accurate data access

Difference Between Data Mining and Data Warehousing

Now, we have understood what is data mining and data warehousing in detail. Here we have listed down the difference between data mining and data warehousing:

Data Mining	Data Warehousing
This procedure involves analysing data patterns	It is exquisitely designed for analytical analysis
Regular data analysis	Periodical data storage
Uses pattern recognition logic to identify patterns	Extracts and stores data to enable easy reporting
It is carried out by business users with the help of engineers	It is carried out by engineers

It helps in extracting data from large data sets	It pools all the relevant data together
Statistics, AI, Machine Learning, and Databases are used in data mining technologies	Integrated, subject-oriented, non-volatile, and time-varying constitute data warehouses
Pattern recognition logic is used for determining patterns	It involves extracting and storing data in perfect order to make efficient reporting
Employs pattern recognition tools to help identify the access patterns	Extracts are stores data in an orderly format, thereby making reporting faster and easier
It helps in creating suggestive patterns of key parameters	When connected with operational business systems like CRM, it deliberately adds value to it

Common Tools and Software Used in Data Warehousing and Data Mining

Let's look at the common tools and software used in data warehousing and data mining: Some of the popular data warehouse tools are:

- Amazon Redshift
- Microsoft Azure
- Google BigQuery
- Snowflake
- Micro Focus Vertica
- Amazon DynamoDB

Some of the popular data mining tools are:

- RapidMiner
- MonkeyLearn
- IBM SPSS Modeler

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- Oracle Data Mining
- Knime
- Weka
- Orange
- H2O
- Apache Mahout
- SAS Enterprise Miner

Common Data Mining and Data Warehousing Techniques

Let's look at the common techniques used in data warehousing vs data mining:

The most common techniques of data mining are:

- Association
- Clustering
- Data Visualisation
- Data Cleaning
- Machine Learning
- Classification
- Neural Networks
- Prediction
- Data Warehousing
- Outlier Detection

Common Data Warehousing Techniques

Some of the most common data warehousing techniques are:

- Database Compression
- Columnar Data Storage
- In-Memory Processing

- Massive Parallel Processing (MPP)

Scope of Data Mining & Data Warehouse

The scope of data mining vs data warehousing is different from each other. Data mining involves sorting enormous data sets to identify relationships and patterns that can easily solve business problems through data analysis. The scope and techniques of data mining enable enterprises to predict future trends and make informed business decisions.

On the other hand, the scope of data warehousing lies within any domain that has something to do with analytics. Now, let us discuss the differences between data mining and data warehousing challenges faced.

Challenges of Data Mining & Data Warehousing

Some of the most common challenges of data mining vs data warehouse challenges:

Some of the most common challenges experienced by data mining are:

- Incomplete and noisy data
- Social and security challenges
- Complex data
- Distributed data
- Efficiency and scalability of algorithms
- Performance
- Incorporating background knowledge
- Improving mining algorithms, etc.

Some of the most common challenges experienced by data warehousing are:

- Manual data processing
- Data quality
- Data Accuracy
- Testing

- Performance, etc.

2.Data Warehousing - Overview

The term "Data Warehouse" was first coined by Bill Inmon in 1990. According to Inmon, a data warehouse is a subject oriented, integrated, time-variant, and non-volatile collection of data. This data helps analysts to take informed decisions in an organization.

An operational database undergoes frequent changes on a daily basis on account of the transactions that take place. Suppose a business executive wants to analyze previous feedback on any data such as a product, a supplier, or any consumer data, then the executive will have no data available to analyze because the previous data has been updated due to transactions.

A data warehouses provides us generalized and consolidated data in multidimensional view. Along with generalized and consolidated view of data, a data warehouses also provides us Online Analytical Processing (OLAP) tools. These tools help us in interactive and effective analysis of data in a multidimensional space. This analysis results in data generalization and data mining.

Data mining functions such as association, clustering, classification, prediction can be integrated with OLAP operations to enhance the interactive mining of knowledge at multiple level of abstraction. That's why data warehouse has now become an important platform for data analysis and online analytical processing.

Understanding a Data Warehouse

- A data warehouse is a database, which is kept separate from the organization's operational database.
- There is no frequent updating done in a data warehouse.
- It possesses consolidated historical data, which helps the organization to analyze its business.
- A data warehouse helps executives to organize, understand, and use their data to take strategic decisions.
- Data warehouse systems help in the integration of diversity of application systems.
- A data warehouse system helps in consolidated historical data analysis.

Why a Data Warehouse is Separated from Operational Databases

A data warehouse is kept separate from operational databases due to the following reasons –

- An operational database is constructed for well-known tasks and workloads such as searching particular records, indexing, etc. In contrast, data warehouse queries are often complex and they present a general form of data.
- Operational databases support concurrent processing of multiple transactions. Concurrency control and recovery mechanisms are required for operational databases to ensure robustness and consistency of the database.
- An operational database query allows to read and modify operations, while an OLAP query needs only **read only** access of stored data.
- An operational database maintains current data. On the other hand, a data warehouse maintains historical data.

Data Warehouse Features

The key features of a data warehouse are discussed below –

- **Subject Oriented** – A data warehouse is subject oriented because it provides information around a subject rather than the organization's ongoing operations. These subjects can be product, customers, suppliers, sales, revenue, etc. A data warehouse does not focus on the ongoing operations, rather it focuses on modelling and analysis of data for decision making.
- **Integrated** – A data warehouse is constructed by integrating data from heterogeneous sources such as relational databases, flat files, etc. This integration enhances the effective analysis of data.
- **Time Variant** – The data collected in a data warehouse is identified with a particular time period. The data in a data warehouse provides information from the historical point of view.
- **Non-volatile** – Non-volatile means the previous data is not erased when new data is added to it. A data warehouse is kept separate from the operational database and therefore frequent changes in operational database is not reflected in the data warehouse.

Note – A data warehouse does not require transaction processing, recovery, and concurrency controls, because it is physically stored and separate from the operational database.

Data Warehouse Applications

As discussed before, a data warehouse helps business executives to organize, analyze, and use their data for decision making. A data warehouse serves as a sole part of a plan-execute-assess "closed-loop" feedback system for the

enterprise management. Data warehouses are widely used in the following fields

–

- Financial services
- Banking services
- Consumer goods
- Retail sectors
- Controlled manufacturing

Types of Data Warehouse

Information processing, analytical processing, and data mining are the three types of data warehouse applications that are discussed below –

- **Information Processing** – A data warehouse allows to process the data stored in it. The data can be processed by means of querying, basic statistical analysis, reporting using crosstabs, tables, charts, or graphs.
- **Analytical Processing** – A data warehouse supports analytical processing of the information stored in it. The data can be analyzed by means of basic OLAP operations, including slice-and-dice, drill down, drill up, and pivoting.
- **Data Mining** – Data mining supports knowledge discovery by finding hidden patterns and associations, constructing analytical models, performing classification and prediction. These mining results can be presented using the visualization tools.

Sr.No.	Data Warehouse (OLAP)	Operational Database(OLTP)
1	It involves historical processing of information.	It involves day-to-day processing.
2	OLAP systems are used by knowledge workers such as executives, managers, and analysts.	OLTP systems are used by clerks, DBAs, or database professionals.
3	It is used to analyze the business.	It is used to run the business.
4	It focuses on Information out.	It focuses on Data in.
5	It is based on Star Schema, Snowflake Schema, and Fact Constellation Schema.	It is based on Entity Relationship Model.
6	It focuses on Information out.	It is application oriented.

7	It contains historical data.	It contains current data.
8	It provides summarized and consolidated data.	It provides primitive and highly detailed data.
9	It provides summarized and multidimensional view of data.	It provides detailed and flat relational view of data.
10	The number of users is in hundreds.	The number of users is in thousands.
11	The number of records accessed is in millions.	The number of records accessed is in tens.
12	The database size is from 100GB to 100 TB.	The database size is from 100 MB to 100 GB.
13	These are highly flexible.	It provides high performance.

3 Differences between Data Warehouse and Database:

Database vs. data warehouse, so what are the main differences between them? Let's take a look at their purpose, use, structure, volume, integration, reporting, analysis, and performance.

Purpose and Use

Database stores structured data in the computer system or software and uses it for the functioning of that particular software or system. On the other hand, a data warehouse aims to store historical and current data for data analysis within a predefined scheme of parameters.

Data Structure

Another difference between a database and a data warehouse is the data structure. Data gathered in a database is structured as efficiently as possible without duplicate info. In contrast, the structure within the data warehouse is denormalized because the data warehouses emphasize the data read over the data writing operations.

Data Volume

A data warehouse volume is much larger than the volume of a database. The main reason for this is that data warehouses are usually used for data mining, while databases are used for the functioning of a particular system.

That's why a data warehouse can process big chunks of data, the Online Analytical Processing (OLAP) data warehouse, while a database has a much lower capacity.

Data Integration

Another difference is the **data integration process**. A database needs written code that'll function like a scheme or a set of data integration parameters, taking into account all the client's applications. The typical database integration scheme is as follows:

- The client sends a request or an application to the database master server.
- The master server checks the request against external and internal sources and extracts it from them.
- The master server consolidates the data.
- The master server serves it back to the customer.

On the contrary, the **data integration process** of data warehouses is much more complex because it combines data from multiple sources into a single unified view. Following are the data integration steps of the **data warehouse integration**:

- New chunks of data enter the system.
- They're checked against the data warehouse parameters and existing integration rules.
- They're combined and stored in the data warehouse.

Query and Reporting

When we compare database vs. data warehouse, we'll see that all databases, except OLTP databases, have limited query and reporting capacities. The reason is that there are historical records within the database. On the other hand, data warehouses store different information with various histories, which makes them ideal for queries and reports.

Data Analysis

The information and data stored in a database have a limited history range, and that's why they're hard to analyze. In contrast, data warehouses are specifically designed for analysis and reporting, so it's much easier to access the data and conduct a query.

Scalability and Performance

Data warehouses are more scalable than databases because they can process large volumes of data. Contrary, databases are more efficient for small-scale queries and transactions.

How to Choose Between Data Warehouse or Database

The database vs. data warehouse debate brings out many differences between the two. However, both have advantages, making it that much harder to choose the right one. To simplify it for you, we've outlined some factors to consider, as well as use cases.

Factors to Consider

To choose dbms vs. data warehouse, you'll have to consider multiple factors. Some essential criteria for making this decision are:

- Available sources
- Storage locations
- Queries and analytics features
- Data types

For example, if you operate with multiple sources, a **data warehouse is a better alternative than a database**. On the other hand, if you want a decentralized storage system and a few locations to store your business-relevant data, a database would be a better option.

As we mentioned, a data warehouse is the best solution for data analytics and queries, while a database is great for application development, accessibility, and real-time information processing.

Use Cases and Scenarios

You can determine whether your business needs a database or a data warehouse based on the various use cases and scenarios. A sheer example of this would be the case when you want to optimize and automate your business process and keep track of customers' info and applications. In this situation, the better option for you would be a database.

Another use case scenario would be evaluating your teams' performance or the success of the various marketing campaigns. Here, the best practice would be to implement a warehouse because it can generate multiple queries and enable thorough data analysis.

Decision-Making Framework

Other crucial factors for choosing the most appropriate system, making a difference between database and data warehouse, are the decision-making frameworks. An example of this would be the OLTP vs. data warehouse frameworks.

Databases use OLTP (Online Transaction Processing), which allows them to delete, change, or update huge numbers within small online transactions. On the contrary, data warehouses usually

use OLAP (Online Analytical Processing, which is efficient for rapidly analyzing huge amounts of data.

Difference between Database and Data Warehouse

4. THE COMPELLING NEED FOR DATA WAREHOUSING

As an information technology professional, you have worked on computer applications as an analyst, programmer, designer, developer, database administrator, or project manager.

Parameter	Database	Data Warehouse
Purpose	Is designed to record	Is designed to analyze
Processing Method	The database uses the Online Transactional Processing (OLTP)	Data warehouse uses Online Analytical Processing (OLAP).
Usage	The database helps to perform fundamental operations for your business	Data warehouse allows you to analyze your business.
Tables and Joins	Tables and joins of a database are complex as they are normalized.	Table and joins are simple in a data warehouse because they are denormalized.
Orientation	Is an application-oriented collection of data	It is a subject-oriented collection of data
Storage limit	Generally limited to a single application	Stores data from any number of applications
Availability	Data is available real-time	Data is refreshed from source systems as and when needed
Usage	ER modeling techniques are used for designing.	Data modeling techniques are used for designing.
Technique	Capture data	Analyze data
Data Type	Data stored in the Database is up to date.	Current and Historical Data is stored in Data Warehouse. May not be up to date.
Storage of data	Flat Relational Approach method is used for data storage.	Data Ware House uses dimensional and normalized approach for the data structure. Example: Star and snowflake schema.
Query Type	Simple transaction queries are used.	Complex queries are used for analysis purpose.
Data Summary	Detailed Data is stored in a database.	It stores highly summarized data.

You have been involved in the design, implementation, and maintenance of systems that support day-to-day business operations. Depending on the industries you have worked in, you must have been involved in applications such as order processing, general ledger, inventory, in-patient billing, checking accounts, insurance claims, and so on.

These applications are important systems that run businesses. They process orders, maintain inventory, keep the accounting books, service the clients, receive payments, and process claims. Without these computer systems, no modern business can survive. Companies started building and using these systems in the 1960s and have become completely dependent on them. As an enterprise grows larger, hundreds of computer applications are needed to support the various business processes. These applications are effective in what they are designed to do. They gather, store, and process all the data needed to successfully perform the daily operations. They provide online information and produce a variety of reports to monitor and run the business. In the 1990s, as businesses grew more complex, corporations spread globally, and competition became fiercer, business executives became desperate for information to stay competitive and improve the bottom line. The operational computer systems did provide information to run the day-to-day operations, but what the executives needed were different kinds of information that could be readily used to make strategic decisions.



Figure 1-1 Organizations' use of data warehousing.

They wanted to know where to build the next warehouse, which product lines to expand, and which markets they should strengthen. The operational systems, important as they were, could not provide strategic information. Businesses, therefore, were compelled to turn to new ways of getting strategic information. Data warehousing is a new paradigm specifically intended to provide vital strategic information.

5.Data warehouse – The building Blocks

Data warehouse is a system used for reporting and data analysis, and it is considered a fundamental component of business intelligence. The **data warehouse building blocks** are the key components that make up this system, and understanding these elements is crucial for anyone looking to implement or optimize a data warehouse.

Data Warehouse Building Blocks

Data Sources

The first of the **data warehouse building blocks** is data sources. These are the various origins from which data is collected. Data can come from internal systems like CRM and ERP systems or external sources like social media, market research, and third-party databases. Identifying and integrating these sources is the foundation of a robust data warehouse.

ETL Process (Extract, Transform, Load)

The ETL process is a critical part of the **data warehouse building blocks**. This process involves extracting data from different sources, transforming it into a format suitable for analysis, and loading it into the data warehouse. ETL ensures that data is clean, consistent, and ready for use.

Data Storage

Data storage is another essential **data warehouse building block**. This component involves the physical storage of data in the data warehouse. Storage solutions can vary from on-premises servers to cloud-based storage systems. The choice of storage depends on factors like data volume, access speed, and security requirements.

Metadata Management

Metadata management is a vital **data warehouse building block** that involves the creation, organization, and maintenance of metadata, which is data about data. Metadata helps in understanding the source, structure, and usage of data within the data warehouse, making it easier to manage and use the data effectively.

Data Modeling

Data modeling is the design phase of the **data warehouse building blocks**. It involves defining the structure of the data warehouse, including the tables, relationships, and data flows. A well-designed data model ensures efficient data retrieval and analysis, making it a crucial component of the data warehouse.

Data Integration

Data integration is the process of combining data from different sources into a unified view. As one of the **data warehouse building blocks**, it ensures that data from disparate systems is brought together in a coherent manner, enabling comprehensive analysis and reporting.

Data Quality

Maintaining high data quality is a fundamental aspect of the **data warehouse building blocks**. This involves ensuring that data is accurate, complete, and reliable. Data quality management includes processes like data cleansing, validation, and verification, which are essential for trustworthy business intelligence.

Data Security

Data security is a non-negotiable **data warehouse building block**. Protecting sensitive data from unauthorized access and ensuring compliance with regulations is critical. This involves implementing security measures such as encryption, access controls, and regular security audits.

User Access Tools

The final **data warehouse building block** is user access tools. These tools enable users to interact with the data warehouse, perform analyses, and generate reports. Tools like SQL queries, business intelligence (BI) software, and data visualization tools are commonly used to access and interpret data.

Why Understanding Data Warehouse Building Blocks Matters

Grasping the **data warehouse building blocks** is essential for building a robust data warehouse that meets your organization's needs. Each building block plays a crucial role in ensuring that data is stored efficiently, remains secure, and is easily accessible for analysis. By understanding these components, businesses can make informed decisions about their data management strategies and tools.

Building a data warehouse is a complex task, but by understanding the **data warehouse building blocks**, you can lay a strong foundation for your data management infrastructure. From data sources and ETL processes to data security and user access tools, each component is vital for creating a system that supports effective data analysis and business intelligence. Invest time in understanding these building blocks to ensure that your data warehouse serves your organization well.

For more insights and detailed guidance on building and optimizing your data warehouse, stay tuned to The Data Group blog. Our experts are here to help you navigate the complexities of data management and achieve your business intelligence goals.

6. Defining Features of Data warehouse

Data warehouses play a crucial role in modern data management and analytics. These centralized data repositories enable organizations to store, manage, and analyze vast amounts of business data from multiple sources. A [data warehouse](#) environment is a centralized repository containing an extraction, transportation, and loading solution,

an [online analytical processing](#) engine, customer analysis tools, and other applications for gathering and delivering information to business users. By understanding the characteristics of data warehouses, businesses can harness their full potential for data-driven decision-making and business intelligence.

Key Characteristics of a Data Warehouse

1. **Subject-Oriented** Data warehouses are designed around specific business subjects such as sales, finance, or customer information. Unlike operational systems, which are process-oriented, data warehouses focus on the analysis of data related to key business areas. This subject-oriented approach allows for more relevant and comprehensive data analysis.
2. **Integrated Data** Integration is a cornerstone of data warehousing. Data from multiple sources, such as relational databases, flat files, and transactional systems, is combined and standardized to provide a unified view. This integration ensures consistency and reliability across the data warehouse, making it a powerful tool for business intelligence.
3. **Time-Variant** Data warehouses store historical data, enabling the analysis of trends and changes over time. This time-variant characteristic is essential for tracking performance, forecasting future trends, and conducting time-based analysis. Data is typically timestamped, allowing for the comparison of data across different periods.
4. **Non-Volatile** Once data is entered into a data warehouse, it is not changed or deleted. This non-volatile nature ensures the integrity and stability of the data, making it reliable for long-term analysis. Historical data remains intact, providing a continuous and accurate historical perspective.
5. **Centralized Repository** A data warehouse acts as a centralized data repository, storing data from various sources in one place. This centralized approach simplifies data management, data access, and data analysis. Business users can access a single source of truth for their data needs, enhancing data consistency and accuracy.
6. **Enterprise Data Warehouse** An enterprise data warehouse (EDW) serves as a centralized data repository for efficient analysis and data access. Utilizing an EDW helps in maintaining data quality, understanding trends, managing large data volumes, and handling user demands. It improves query performance, maintains historical data, and stores heterogeneous data. Additionally, an EDW provides functions such as restructuring data, adding value to operational business applications, merging data for a common data model, and using better Business Intelligence (BI) analytics.

Data Warehousing Process

The data warehousing process involves several key steps to ensure that data is accurately collected, integrated, and stored for analysis. These steps include:

1. **Data Extraction, Transformation, and Load (ETL)** ETL is a critical process in data warehousing. Data is extracted from multiple sources, transformed into a consistent format, and loaded into the data warehouse. This process ensures that data is cleaned, standardized, and ready for analysis. [ETL tools](#) automate this process, improving efficiency and accuracy.
2. **Data Cleaning** Data cleaning is the process of identifying and correcting errors, inconsistencies, and inaccuracies in the data. This step is essential to ensure data quality

and reliability. Data cleaning involves removing duplicates, correcting errors, and standardizing data formats.

3. **Data Integration** Data integration involves combining data from multiple sources into a unified view. This process ensures that data from different sources is consistent and reliable. Data integration tools help automate this process, making it easier to manage and integrate data from various sources.
4. **Data Storage** Data storage is a key component of data warehousing. Data warehouses store data in a structured format, typically using relational databases. This structured storage enables efficient data retrieval and analysis. Cloud data warehouses provide scalable and flexible storage solutions, making it easier to manage large volumes of data.
5. **Data Access** Data access involves allowing business users to query and analyze data in the data warehouse. Online analytical processing ([OLAP](#)) tools and reporting tools enable users to perform complex queries, generate reports, and gain insights from the data. These tools provide a user-friendly interface for accessing and analyzing data.

Data Warehouse Architecture

The architecture of a data warehouse consists of several key components, each playing a vital role in the overall functionality and performance of the system. The main [components of a data warehouse](#) architecture include:

1. **Data Sources** Data sources are the systems and databases from which data is extracted for the data warehouse. These sources can include relational databases, flat files, transactional systems, and other operational systems. The ETL process integrates Data from these sources into the data warehouse.
2. **ETL Layer** The ETL layer is responsible for extracting data from various sources, transforming it into a consistent format, and loading it into the data warehouse. This layer ensures that data is cleaned, standardized, and integrated before being stored in the data warehouse.
3. **Data Storage Layer** The data storage layer is where data is stored in the data warehouse. This layer typically uses relational databases to store structured data. Cloud data warehouses offer scalable storage solutions, allowing organizations to manage large volumes of data efficiently.
4. **Data Access Layer** The data access layer provides business users with the tools and interfaces to query and analyze data in the data warehouse. This layer includes OLAP tools, reporting tools, and other [data analysis tools](#) that enable users to gain insights from the data.
5. **Metadata Layer** The metadata layer stores information about the data in the data warehouse, including data definitions, data mappings, and data lineage. Metadata helps users understand the structure and meaning of the data, making it easier to navigate and analyze the data warehouse.
6. **Data Mart Layer** Data marts are specialized subsets of the data warehouse, designed to meet the specific needs of individual business units or departments. Data marts provide a focused view of the data, allowing for more targeted analysis and reporting.

Benefits of Data Warehousing

Data warehousing offers numerous benefits to organizations, enabling them to make more informed decisions and improve their business processes. Some of the key benefits include:

1. **Improved Data Quality** Data warehousing processes, such as data cleaning and integration, ensure that data is accurate, consistent, and reliable. This improved data quality enhances the accuracy of analysis and decision-making.
2. **Enhanced Business Intelligence** Data warehouses provide a centralized repository of data, enabling more comprehensive and in-depth analysis. Business intelligence tools can leverage this data to generate insights, identify trends, and support data-driven decision-making.
3. **Scalable and Flexible Storage** Cloud data warehouses offer scalable storage solutions, allowing organizations to manage large volumes of data efficiently. This scalability ensures that the data warehouse can grow with the organization's needs, providing flexible storage options.
4. **Efficient Data Access** Data warehouses provide efficient data access, enabling business users to query and analyze data quickly and easily. OLAP tools and reporting tools provide a user-friendly interface for accessing and analyzing data, improving efficiency and productivity.
5. **Historical Data Analysis** Data warehouses store historical data, allowing organizations to analyze trends and changes over time. This historical perspective is essential for tracking performance, forecasting future trends, and conducting time-based analysis.
6. **Centralized Data Management** Data warehouses provide a centralized repository for managing data from multiple sources. This centralized approach simplifies data management, ensuring data consistency and accuracy. Business users can access a single source of truth for their data needs.

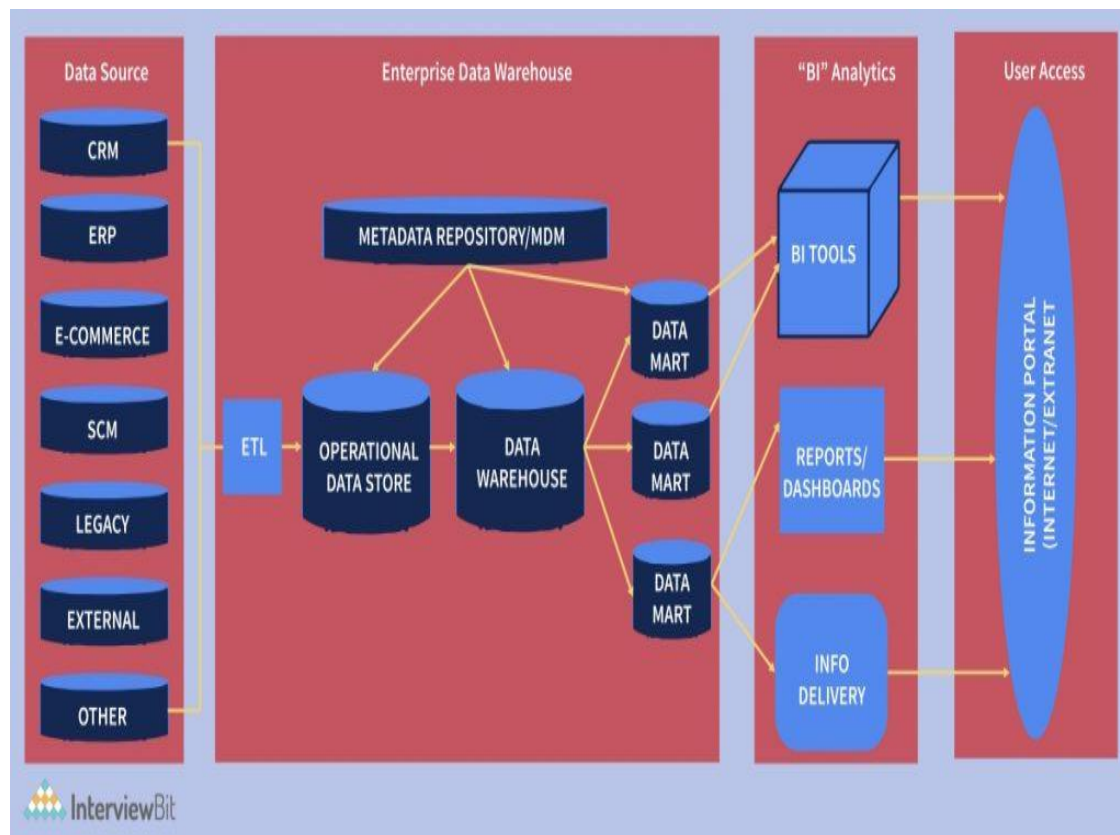
7. Difference between Data Warehouse and Data Mart

- Both **Data Warehouse** and **Data Mart** are used for store the data. The main difference between Data warehouse and Data mart is that, **Data Warehouse** is the type of database which is **data-oriented** in nature. While, **Data Mart** is the type of database which is the **project-oriented** in nature. The other difference between these two the Data warehouse and the Data mart is that, Data warehouse is large in scope whereas Data mart is limited in scope. Let's see the difference between Data warehouse and Data mart:

Sl.No	Data Warehouse	Data Mart
1.	Data warehouse is a Centralised system.	While it is a decentralised system.
2.	In data warehouse, lightly denormalization takes place.	While in Data mart, highly denormalization takes place.
3.	Data warehouse is top-down model.	While it is a bottom-up model.
4.	To built a warehouse is difficult.	While to build a mart is easy.
5.	In data warehouse, Fact constellation schema is used.	While in this, Star schema and snowflake schema are used.
6.	Data Warehouse is flexible.	While it is not flexible.
7.	Data Warehouse is the data-oriented in nature.	While it is the project-oriented in nature.
8.	Data Ware house has long life.	While data-mart has short life than warehouse.
9.	In Data Warehouse, Data are contained in detail form.	While in this, data are contained in summarized form.
10.	Data Warehouse is vast in size.	While data mart is smaller than warehouse.

Sl.No	Data Warehouse	Data Mart
11.	The Data Warehouse might be somewhere between 100 GB and 1 TB+ in size.	The Size of Data Mart is less than 100 GB.
12.	The time it takes to implement a data warehouse might range from months to years.	The Data Mart deployment procedure is time-limited to a few months.
13.	It uses a lot of data and has comprehensive operational data.	Operational data are not present in Data Mart.
14.	It collects data from various data sources.	It generally stores data from a data warehouse.
15.	Long time for processing the data because of large data.	Less time for processing the data because of handling only a small amount of data.
16.	Complicated design process of creating schemas and views.	Easy design process of creating schemas and views.

8. Data Warehouse Components



Data Warehouse Database

The foremost element of a Data Warehouse architecture is a database that stores all company data and makes it manageable to report.. This implies you ought to pick which type of database you'll use to stow data in your warehouse.

Mentioned below are the four database types that you can utilize:

- **Typical relational databases:** You perhaps use these row-centred databases on an everyday basis, for instance, SAP, Oracle, Microsoft SQL Server, and IBM DB2.
- **Analytics databases** are specifically designed for storing data to support and handle analytics, like Greenplum and Teradata.
- **Data warehouse applications** aren't exactly storage databases, but lots of dealers now provide applications that offer software for data management along with hardware for data storage. For instance, Oracle Exadata, IBM Netezza, and SAP Hana.

- **Cloud-based databases** can be retrieved and hosted on the cloud so that you don't need to obtain any hardware to set up your data warehouse—for instance, Google BigQuery, Microsoft Azure SQL and Amazon Redshift.

Sourcing, Acquisition, Cleanup, and Transformation Tools (ETL)

A substantial part of the implementation effort is spent pulling data from operational systems and putting it in a format appropriate for informational applications that operate off the data warehouse. The data sourcing, cleanup, migration, and transformation tool fulfill all of the summarizations, conversions, structural changes, key changes, and condensations required to convert disparate data into data that can be utilized by the decision support tool. They produce the programs and control statements, such as the MVS job-control language (JCL), UNIX scripts, SQL data definition language (DDL), and COBOL programs, required to transfer data into the data warehouse for numerous operational systems. Metadata is also maintained by these tools. The functionality includes:

- Erasing undesirable data from operational databases
- Transforming to common data definitions and names
- Setting defaults for missing data
- Accommodating source data definition modifications
- Anonymize data as per regulatory stipulations.
- Eliminating unwanted data in operational databases from loading into Data warehouse.
- Search and replace common names and definitions for data arriving from different sources.
- Calculating summaries and derived data
- In case of missing data, populate them with defaults.
- De-duplicated repeated data arriving from multiple datasources.

The data sourcing, cleanup, extract, transformation, and migration tools have to deal with challenges of Database & Data heterogeneity

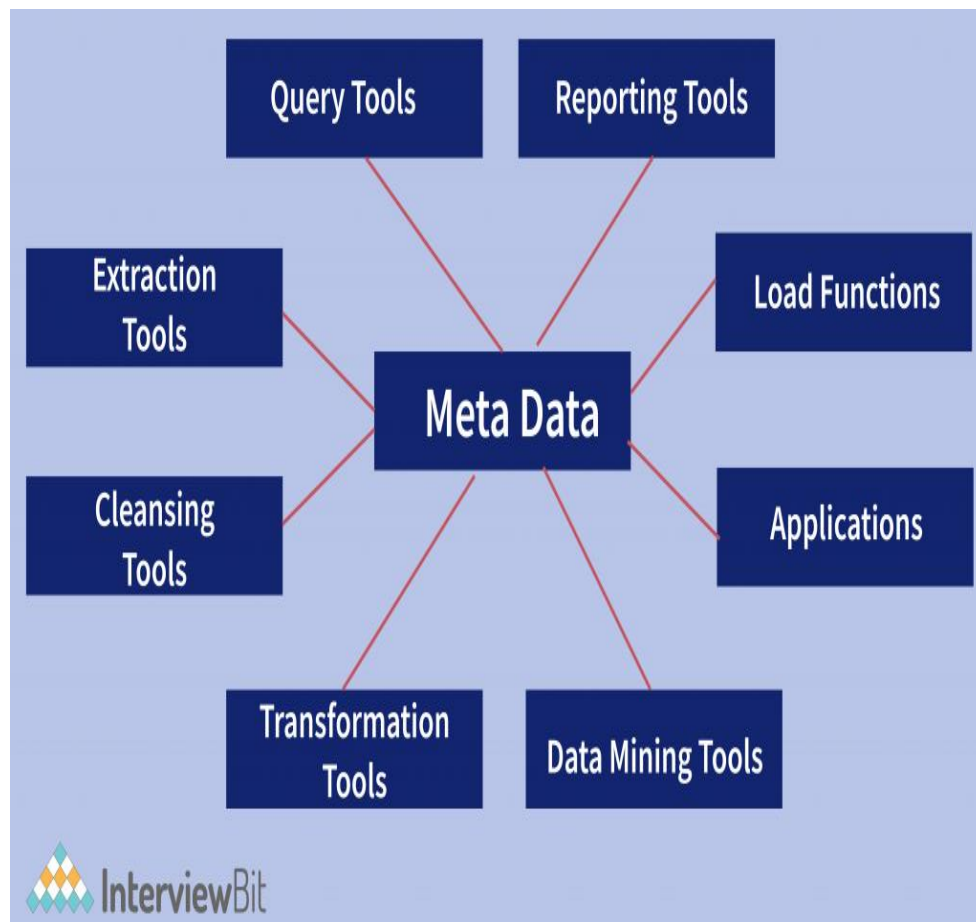
- Database heterogeneity. DBMSs are very diverse in data access language, data models, concurrency, data navigation, operations, recovery, integrity, etc.
- Data heterogeneity. This is the distinction in the way data is depicted and utilized in various models – synonyms, homonyms, unit compatibility (the

U.S. vs metric), diverse attributes for the identical entity, and different ways of modelling similar fact.

A considerable amount of time and effort is saved with these tools. Nevertheless, substantial drawbacks can be seen. For instance, numerous available tools are generally good for easier data extracts.

Often, customized extract routines ought to be developed for the more complex data extraction methods.

Metadata



Metadata defines the data warehouse database and presents a framework for data, in the data warehouse architecture. It assists in handling, preserving,

constructing, and making use of the data warehouse. Metadata helps to answer the following questions

What tables, attributes, and keys does the Data Warehouse contain?

Where did the data come from?

How many times do data get reloaded?

What transformations were applied with cleansing?

In data warehousing, there are two kinds of metadata:

- **Technical Metadata** includes data that can be utilized by programmers and managers when performing administration tasks and warehouse development.
- **Business Metadata** contains data that shows a readily comprehensible standpoint of the information stowed in the warehouse.

Metadata has an essential role to play for companies and technical teams to comprehend the information present in the warehouse and transform it into information. Your data warehouse is a process and isn't a project. To make your performance as efficacious as possible, you ought to, indeed take an agile approach, which entails a metadata-driven data warehouse architecture. This is a visual method to data warehousing that supports metadata-enriched data models to operate each element of the development process from noting source systems to copying schemas in a physical database and enabling mapping from source to destination.

At the metadata level, the data warehouse schema is set up, which implies you don't have to get worked up regarding code quality and how it will meet high volumes of data. You can control and manage your data without going into the code. Furthermore, before deploying and replicating your schema in any leading database, you can concurrently test data warehouse models. A metadata-driven method expedites an iterative development culture and your data warehouse deployment is future-proofed. Hence, you can modify the current infrastructure with the new prerequisites without disturbing your data warehouse's usability and integrity. Paired with automation capabilities, a metadata-driven data warehouse architecture can simplify deployment, design, and development, causing a rich data warehouse implementation.

Query Tools

The main intent of data warehousing is to deliver data to business users for the purpose of strategic decision-making. Utilizing front-end tools, these users interact with the data warehouse. A few of these tools need an information specialist, though many end users acquire specialization in the tools. Tools are categorized into four types: application development tools, query and reporting tools, data mining tools, and online analytical processing tools

- Query and Reporting tools can be segregated into two levels: reporting tools and managed query tools. Additionally, reporting tools can be split into report writers and production reporting tools.
 - Production reporting tools allow businesses to develop regular operational reports or back high-volume batch jobs including computing and printing paychecks.
 - On the other hand, Report writers are affordable desktop tools developed for end-users.
- Managed query tools to protect end users from the intricacies of SQL and database structures by adding a meta-layer between the database and users. These tools are created for ease of use, point-and-click operations that accept SQL or develop SQL database queries.

Usually, the analytical requirements of the data warehouse user community surpass the built-in capacities of reporting and query tools. In these circumstances, companies usually rely on the tried-and-true approach of in-house application development with the help of graphical development conditions like Visual Basic, PowerBuilder, and Forte. These application development platforms blend well with common OLAP tools and access all significant database systems such as Sybase, Informix, and Oracle.

- OLAP tools are based on the notions of dimensional data models and connected databases, and let users study the data utilizing multidimensional and elaborate views. Standard business applications contain the effectiveness of a sales program or marketing campaign, product profitability and performance, capacity planning, and sales forecasting. These tools believe that the data is managed in a multidimensional model.

Today, a crucial success element for any enterprise is the capability to utilize data efficiently. The process of locating significant new correlations, trends, and

patterns by exploring huge chunks of data stowed in the warehouse utilizing statistical and mathematical techniques and artificial intelligence is data mining.

Data Marts

The notion of a data mart is generating dollops of excitement and draws considerable attention in the data warehouse industry. Mainly, data marts are offered as an alternative to a data warehouse that takes very little time and money to create. Nevertheless, data mart implies various things to various people. Another meaning of it is a data store, the subsidiary of a data warehouse of integrated data. The data mart is directed at a division of data, referred to as a subject area that is designed for the benefit of a dedicated set of users. It might be a collection of summarized, denormalized, or aggregated data. Oftentimes, such a set could be situated in the data warehouse rather than a physically distinct store of data. However, in certain cases, the data mart is a physically distinct store of data and resides on a separate database server, oftentimes a local area network serving a certain user group. At times, the data mart just includes relational OLAP technology, which makes a highly denormalized dimensional model, for instance, star schema, executed on a relational database. The resultant hypercubes of information are utilized for research by users having a shared interest in a restricted part of the database. These kinds of data marts are known as dependent data marts as their data is received from the data warehouse, and maintains a high value since various users are all accessing the data views received from the single integrated version of the data, no matter the way they are deployed and how many other enabling technologies are utilized.

Sadly, the ambiguous statements regarding the simplicity and lower price of data marts occasionally lead to organizations wrongly placing them as an alternative to the data warehouse. This perspective describes independent data marts that in fact, portray fragmented point solutions to a spectrum of business issues in the enterprise. This kind of undertaking should be seldom deployed in the context of a general technology or applications architecture. Certainly, it is missing the element that is at the core of the data warehousing notion — that of data integration. Every self-reliant data mart makes its inferences regarding how to capsulize the data, and the data across different data marts may not be uniform. Also, the notion of an autonomous data mart is precarious — just as the first data mart is made, other institutions, subject areas, and groups within the enterprise venture on the job of creating their data marts. Therefore, you build an environment where numerous operational systems provide numerous non-integrated data marts that are

frequently overlapping in job scheduling, data content, management, and connectivity. In a nutshell, you have converted a complicated many-to-one issue of creating a data warehouse from external and operational data sources to a many-to-many management and sourcing nightmare.

Data Warehouse Management and Administration

Data warehouses appear to be as much as 4 times as extensive as related operational databases, almost terabytes in size, which depends on how much history requires to be preserved. In real-time, they are not synchronized to the associated operational data but are revised as frequently as once a day, considering the needs of the application. Additionally, almost all products of data warehouses such as gateways can clearly access numerous enterprise data sources without the need to rewrite applications to analyze and use the data. Also, in a heterogeneous data warehouse environment, the different databases live on disparate systems, which requires inter-networking tools. The necessity to handle this environment is obvious. Handling data warehouses such as security and priority management, data quality checks, tracking updates from multiple sources, auditing and reporting data warehouse usage and status, handling and revising metadata, purging data, copying, subsetting, and distributing data, backup and retrieval, and data warehouse storage management.

Information Delivery System

The data delivery component is utilized to allow the method of subscribing for data warehouse data and getting it delivered to one or more places as per a certain user-specified scheduling algorithm. The information delivery system allocates warehouse-stored data and additional information objects to different data warehouses and end-user products like local databases and spreadsheets. Delivery of data can be based on the time of day or the fulfilment of an external event. The idea for the delivery systems component is made based on the fact that once the data warehouse is set and functional, its users will not have to be cognizant of its maintenance and location. All they require is a report or an analytical thesis of data at an exact point in time. With the expansion of the World Wide Web and the Internet, such a delivery system may support the benefit of the Internet by providing warehouse-enabled information to several end-users through the universal worldwide network.

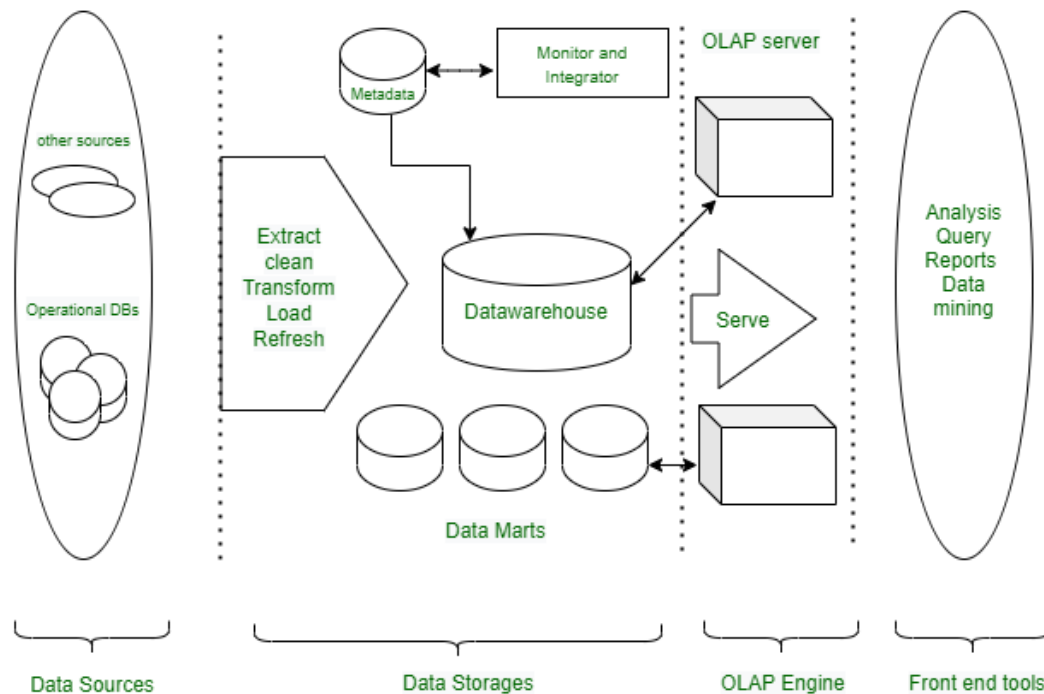
9. Three Tier Data Warehouse Architecture

Multi-tier architecture of Data Warehouse

A data warehouse is Representable by data integration from multiple heterogeneous sources. It was defined by **Bill Inmon** in 1990. The data warehouse is an integrated, subject-oriented, time-variant, and non-volatile collection of data. A Data Warehouse is structured by data integration from multiple heterogeneous sources. It is a system used for data analysis and reporting. A data warehouse is deliberate a core factor of business intelligence. BI technology provides a historical, current, and predictive view of business operations without data mining many businesses may not be able to perform effective market analysis, the strength and weakness of their competitors, profitable decisions, etc.

Data Warehouse is referred to the data repository that is maintained separately from the organization's operational data. **Multi-Tier Data Warehouse Architecture consists of the following components:**

1. Bottom Tier
2. Middle Tier
3. Top Tier



Three/Multi-tier Architecture of Data Warehouse

Bottom Tier(Data sources and data storage) :

1. The bottom Tier usually consists of Data Sources and Data Storage.
2. It is a warehouse database server. For Example RDBMS.
3. In Bottom Tier, using the application program interface(called gateways), data is extracted from operational and external sources.
4. Application Program Interface likes ODBC(Open Database Connection), OLE-DB(Open-Linking and Embedding for Database), JDBC(Java Database Connection) is supported.
5. ETL stands for Extract, Transform, and Load.

Several popular ETL tools include:

- I. IBM Infosphere
- II. Informatica
- III. Confluent
- IV. Microsoft SSIS
- V. Snaplogic
- VI. Alooma

Middle Tier :

The middle tier is an OLAP server that is typically implemented using either :
A relational OLAP (ROLAP) model (i.e., an extended relational DBMS that maps operations from standard data to standard data);

or

A multidimensional OLAP (MOLAP) model (ie, a special purpose server that directly implements multidimensional data and operations).

OLAP server models come in three different categories, including:

1. **ROLAP:** A relational database is not converted into a multidimensional database; rather, a relational database is actively broken down into several dimensions as part of relational online analytical processing(ROLAP). This is used when everything that is contained in the repository is a relational database system.
2. **MOLAP:** A different type of online analytical processing called multidimensional online analytical processing(MOLAP) includes directories and catalogs that are immediately integrated into its multidimensional database system. This is used when all that is contained in the repository is the multidimensional database system.
3. **HOLAP:** A combination of relational and multidimensional online analytical processing paradigms is hybrid online analytical processing(HOLAP). HOLAP is the ideal option for a seamless functional flow across the database systems when the repository houses both the relational database management system and the multidimensional database management system.

Top Tier :

The top tier is a front-end client layer, which includes query and reporting tools, analysis tools, and/or data mining tools (eg, trend analysis, prediction, etc.).

Here are a few Top Tier tools that are often used:

- I. SAP BW
- II. SAS Business Intelligence
- III. IBM Cognos
- IV. Crystal Reports
- V. Microsoft BI Platform

Data Warehouse Models :

From the architecture point of view, there are three warehouse models-

Enterprise Warehouse:-

- An enterprise warehouse collects all information topics spread throughout the organization.
- It provides corporate-wide data integration, typically from one or several operational systems or external information providers, and is cross-functional in scope.
- It usually contains detailed data as well as summarized data and can range in size from a few gigabytes to hundreds of gigabytes, terabytes, or beyond. Can be an enterprise data warehouse.

- The traditional mainframe, computer super server, or parallel architecture has been implemented on platforms. This requires extensive commercial modeling and may take years to design and manufacture.

Data Mart:-

- A data mart contains a subset of corporate-wide data that is important to a specific group of users.
- The scope is limited to specific selected subjects.
- For example, a marketing data mart may limit its topics to customers, goods, and sales.
- The data contained in the data marts are summarized. Data marts are typically applied to low-cost departmental servers that are Unix/Linux or Windows-based.
- The implementation cycle of a data mart is more likely to be measured in weeks rather than months or years. However, it can be in the long run, complex integration is involved in its design and planning were not enterprise-wide.

Virtual Warehouse:-

- A virtual warehouse is a group of views on an operational database.
- For efficient query processing, only a few possible summary views can be physical.
- Creating a virtual warehouse is easy, but requires additional capacity on operational database servers.

Advantages of Multi-Tier Architecture of Data warehouse

1. **Scalability:** Various components can be added, deleted, or updated in accordance with the data warehouse's shifting needs and specifications.
2. **Better Performance:** The several layers enable parallel and efficient processing, which enhances performance and reaction times.
3. **Modularity:** The architecture supports modular design, which facilitates the creation, testing, and deployment of separate components.
4. **Security:** The data warehouse's overall security can be improved by applying various security measures to various layers.
5. **Improved Resource Management:** Different tiers can be tuned to use the proper hardware resources, cutting expenses overall and increasing effectiveness.
6. **Easier Maintenance:** Maintenance is simpler because individual components can be updated or maintained without affecting the data warehouse as a whole.
7. **Improved Reliability:** Using many tiers can offer redundancy and failover capabilities, enhancing the data warehouse's overall reliability.

10. Data Warehousing - Metadata Concepts

What is Metadata?

Metadata is simply defined as data about data. The data that is used to represent other data is known as metadata. For example, the index of a book serves as a metadata for the contents in the book. In other words, we can say that metadata is the summarized data that leads us to detailed data. In terms of data warehouse, we can define metadata as follows.

- Metadata is the road-map to a data warehouse.
- Metadata in a data warehouse defines the warehouse objects.
- Metadata acts as a directory. This directory helps the decision support system to locate the contents of a data warehouse.

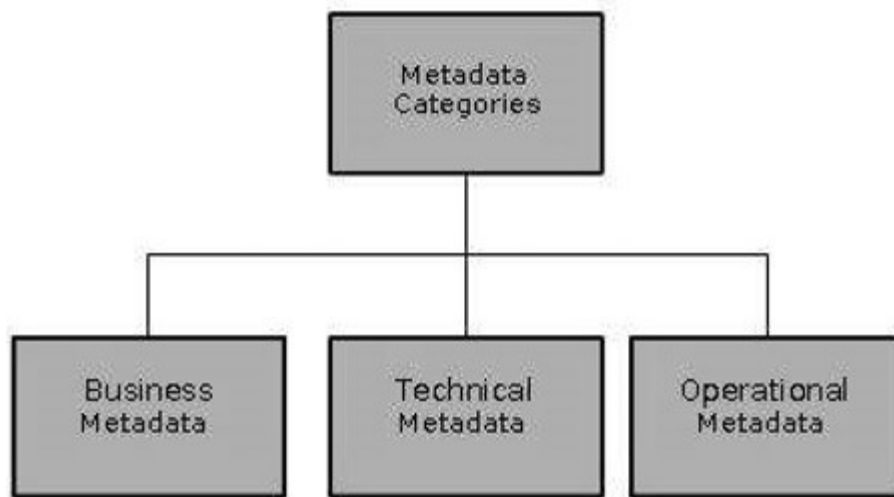
Note – In a data warehouse, we create metadata for the data names and definitions of a given data warehouse. Along with this metadata, additional metadata is also created for time-stamping any extracted data, the source of extracted data.

Categories of Metadata

Metadata can be broadly categorized into three categories –

- **Business Metadata** – It has the data ownership information, business definition, and changing policies.
- **Technical Metadata** – It includes database system names, table and column names and sizes, data types and allowed values. Technical metadata also includes structural information such as primary and foreign key attributes and indices.

- **Operational Metadata** – It includes currency of data and data lineage. Currency of data means whether the data is active, archived, or purged. Lineage of data means the history of data migrated and transformation applied on it.



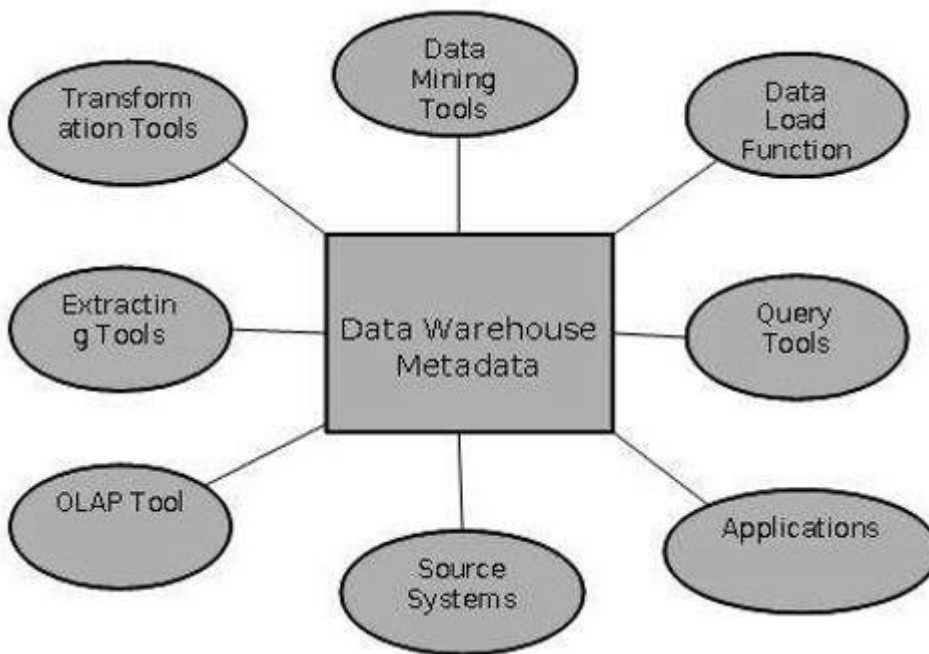
Role of Metadata

Metadata has a very important role in a data warehouse. The role of metadata in a warehouse is different from the warehouse data, yet it plays an important role. The various roles of metadata are explained below.

- Metadata acts as a directory.
- This directory helps the decision support system to locate the contents of the data warehouse.
- Metadata helps in decision support system for mapping of data when data is transformed from operational environment to data warehouse environment.
- Metadata helps in summarization between current detailed data and highly summarized data.
- Metadata also helps in summarization between lightly detailed data and highly summarized data.
- Metadata is used for query tools.
- Metadata is used in extraction and cleansing tools.

- Metadata is used in reporting tools.
- Metadata is used in transformation tools.
- Metadata plays an important role in loading functions.

The following diagram shows the roles of metadata.



Metadata Repository

Metadata repository is an integral part of a data warehouse system. It has the following metadata –

- **Definition of data warehouse** – It includes the description of structure of data warehouse. The description is defined by schema, view, hierarchies, derived data definitions, and data mart locations and contents.
- **Business metadata** – It contains has the data ownership information, business definition, and changing policies.
- **Operational Metadata** – It includes currency of data and data lineage. Currency of data means whether the data is active, archived, or purged. Lineage of data means the history of data migrated and transformation applied on it.

- **Data for mapping from operational environment to data warehouse** – It includes the source databases and their contents, data extraction, data partition cleaning, transformation rules, data refresh and purging rules.
- **Algorithms for summarization** – It includes dimension algorithms, data on granularity, aggregation, summarizing, etc.

Challenges for Metadata Management

The importance of metadata can not be overstated. Metadata helps in driving the accuracy of reports, validates data transformation, and ensures the accuracy of calculations. Metadata also enforces the definition of business terms to business end-users. With all these uses of metadata, it also has its challenges. Some of the challenges are discussed below.

- Metadata in a big organization is scattered across the organization. This metadata is spread in spreadsheets, databases, and applications.
- Metadata could be present in text files or multimedia files. To use this data for information managementsolutions, it has to be correctly defined.
- There are no industry-wide accepted standards. Data management solution vendors have narrow focus.
- There are no easy and accepted methods of passing metadata.

Why Do We Need a Data Mart?

Listed below are the reasons to create a data mart –

- To partition data in order to impose **access control strategies**.
- To speed up the queries by reducing the volume of data to be scanned.
- To segment data into different hardware platforms.
- To structure data in a form suitable for a user access tool.

Note – Do not data mart for any other reason since the operation cost of data marting could be very high. Before data marting, make sure that data marting strategy is appropriate for your particular solution.

Cost-effective Data Marting

Follow the steps given below to make data marting cost-effective –

- Identify the Functional Splits
- Identify User Access Tool Requirements
- Identify Access Control Issues

Identify the Functional Splits

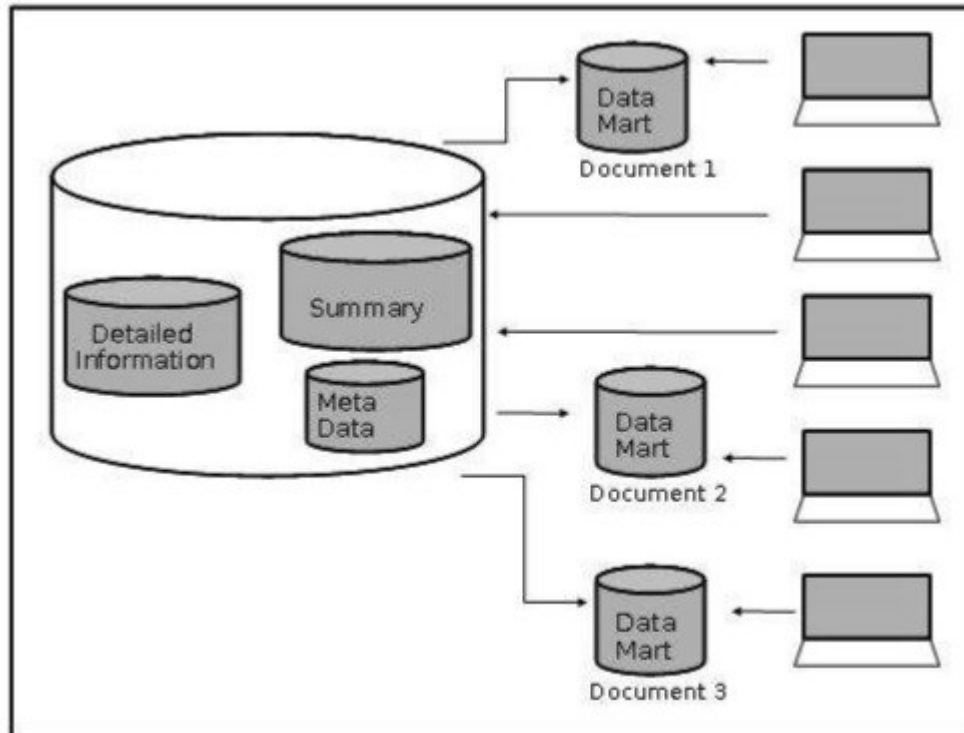
In this step, we determine if the organization has natural functional splits. We look for departmental splits, and we determine whether the way in which departments use information tend to be in isolation from the rest of the organization. Let's have an example.

Consider a retail organization, where each merchant is accountable for maximizing the sales of a group of products.

For this, the following are the valuable information –

- sales transaction on a daily basis
- sales forecast on a weekly basis
- stock position on a daily basis
- stock movements on a daily basis

As the merchant is not interested in the products they are not dealing with, the data marting is a subset of the data dealing with the product group of interest. The following diagram shows data marting for different users.



Given below are the issues to be taken into account while determining the functional split –

- The structure of the department may change.
- The products might switch from one department to other.
- The merchant could query the sales trend of other products to analyze what is happening to the sales.

Note – We need to determine the business benefits and technical feasibility of using a datamart.

Identify User Access Tool Requirements

We need data marts to support **user access tools** that require internal data structures. The data insuch structures are outside the control of data warehouse but need to be populated and updated on a regular basis.

There are some tools that populate directly from the source system but some cannot. Therefore additional requirements outside the scope of the tool are needed to be identified for future.

Note – In order to ensure consistency of data across all access tools, the data should not be directly populated from the data warehouse, rather each tool must have its own data mart.

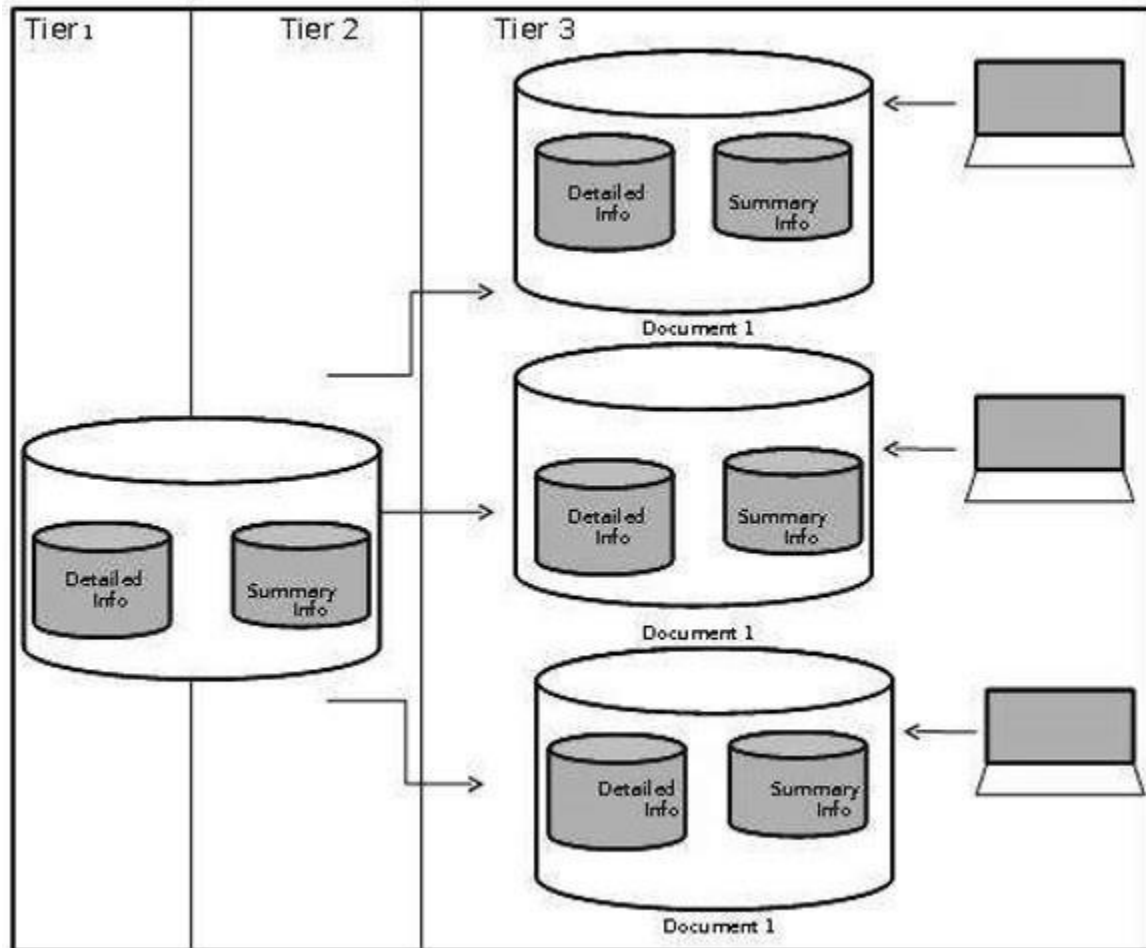
Identify Access Control Issues

There should be privacy rules to ensure the data is accessed by authorized users only. For example a data warehouse for retail banking institution ensures that all the accounts belong to the same legal entity. Privacy laws can force you to totally prevent access to information that is not owned by the specific bank.

Data marts allow us to build a complete wall by physically separating data segments within the data warehouse. To avoid possible privacy problems, the detailed data can be removed from the data warehouse. We can create data mart for each legal entity and load it via data warehouse, with detailed account data.

Designing Data Marts

Data marts should be designed as a smaller version of starflake schema within the data warehouse and should match with the database design of the data warehouse. It helps in maintaining control over database instances.



The summaries are data marted in the same way as they would have been designed within the data warehouse. Summary tables help to utilize all dimension data in the starflake schema.

Cost of Data Marting

The cost measures for data marting are as follows –

- Hardware and Software Cost
- Network Access
- Time Window Constraints

Hardware and Software Cost

Although data marts are created on the same hardware, they require some additional hardware and software. To handle user queries, it requires additional processing power and disk storage. If detailed data and the data mart exist within the data warehouse, then we would face additional cost to store and manage replicated data.

Note – Data marting is more expensive than aggregations, therefore it should be used as an additional strategy and not as an alternative strategy.

Network Access

A data mart could be on a different location from the data warehouse, so we should ensure that the LAN or WAN has the capacity to handle the data volumes being transferred within the **data mart load process**.

Time Window Constraints

The extent to which a data mart loading process will eat into the available time window depends on the complexity of the transformations and the data volumes being shipped. The determination of how many data marts are possible depends on –

- Network capacity.
- Time window available
- Volume of data being transferred
- Mechanisms being used to insert data into a data mart

11. **Data Cleaning and Pre-processing**

Data preprocessing involves the transformation of the raw dataset into an understandable format. Preprocessing data is a fundamental stage in data mining to improve data efficiency. The data preprocessing methods directly affect the outcomes of any analytic algorithm.

Data preprocessing is generally carried out in 7 simple steps:

Steps In Data Preprocessing:

1. Gathering the data
2. Import the dataset & Libraries
3. Dealing with Missing Values
4. Divide the dataset into Dependent & Independent variable
5. dealing with Categorical values
6. Split the dataset into training and test set
7. Feature Scaling

1. Gathering the data

Data is raw information, its the representation of both human and machine observation of the world. Dataset entirely depends on what

type of problem you want to solve. Each problem in machine learning has its own unique approach.

Here i am sharing some website with you to get the dataset :

1. Kaggle: Kaggle is my personal favorite one to get the dataset.
<https://www.kaggle.com/datasets>
2. UCI Machine Learning Repository: One of the oldest sources on the web to get the dataset.
<http://mlr.cs.umass.edu/ml/>
3. This awesome GitHub repository has high-quality datasets.
<https://github.com/awesomedata/awesome-public-datasets>
4. And if you are looking for Government's Open Data then here is few of them:
Indian Government: <http://data.gov.in>
US Government: <https://www.data.gov/>
British Government: <https://data.gov.uk/>
France Government: <https://www.data.gouv.fr/en/>

2. Import the dataset & Libraries

First step is usually importing the libraries that will be needed in the program. A library is essentially a collection of modules that can be called and used.

And can be import the libraries in python code with the help of *'import'* keyword.

```
In [1]: 1 import numpy as np
        2 import pandas as pd
```

Importing the dataset

Loading the data using Pandas library using `read_csv()` method.

```
In [2]: 1 # Importing Dataset
        2 dataset = pd.read_csv('Data.csv')
        3 dataset
```

```
Out[2]:
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	Germany	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Here we have data in csv format, there is many kind of file can be read by using pandas library as shown below:

```
In [ ]: 1 pd.rea
        read_clipboard
In [ ]: 1 read_csv
        read_excel
In [ ]: 1 read_feather
        read_fwf
In [ ]: 1 read_gbq
        read_hdf
In [ ]: 1 read_html
        read_json
In [ ]: 1 read_msgpack
```

3. Dealing with Missing Values

Sometimes we may find some data are missing in the dataset. if we found then we will remove those rows or we can calculate either **mean, mode or median** of the feature and replace it with

missing values. This is an approximation which can add variance to the dataset.

#Check for null values:

we can check the null values in our dataset with pandas library as below.

```
In [12]: 1 dataset.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
Country      10 non-null object
Age          9 non-null float64
Salary       9 non-null float64
Purchased    10 non-null object
dtypes: float64(2), object(2)
memory usage: 400.0+ bytes
```

With the help of *info()* we can found total number of entries as well as count of non-null values with datatype of all features.

we also can use *dataset.isna()* to see the of null values in our dataset.

```
In [13]: 1 dataset.isna()

Out[13]:
```

	Country	Age	Salary	Purchased
0	False	False	False	False
1	False	False	False	False
2	False	False	False	False
3	False	False	False	False
4	False	False	True	False
5	False	False	False	False
6	False	True	False	False
7	False	False	False	False
8	False	False	False	False
9	False	False	False	False

But usually we work on large dataset so it will be a good thing to get the count of all null values corresponding to each features and it will be done by using *sum()*.

```
In [14]: 1 # Tacking care of null values
         2 dataset.isna().sum()
```

```
Out[14]: Country      0
         Age          1
         Salary       1
         Purchased    0
         dtype: int64
```

As we can see ‘**Age**’ and ‘**Salary**’ containing null values.

#Drop Null values:

Pandas provide a **dropna()** function that can be used to drop either row or columns with missing data. We can use *dropna()* to remove all the rows with missing data.

```
In [15]: 1 dataset_1 = dataset.dropna()
```

```
In [16]: 1 dataset_1
```

```
Out[16]:
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
5	France	35.0	58000.0	Yes
7	France	48.0	79000.0	Yes
8	Germany	50.0	83000.0	No
9	France	37.0	67000.0	Yes

We can see before there is 4th and 6th index have null values. Now as above we can see both rows with missing data has been removed. But this is **not always** a good idea. Sometime we have small dataset, as we used in our example and removing the whole row

means somewhere we are deleting some valuable information from dataset.

#Replacing Null values with Strategy:

For replacing null values we use the strategy that can be applied on a feature which has numeric data. We can calculate the *Mean*, *Median* or *Mode* of the feature and replace it with the missing values.

```
In [19]: 1 dataset.fillna(dataset.mean(), inplace=True)
```

```
In [20]: 1 dataset
```

Out[20]:

	Country	Age	Salary	Purchased
0	France	44.000000	72000.000000	No
1	Spain	27.000000	48000.000000	Yes
2	Germany	30.000000	54000.000000	No
3	Spain	38.000000	61000.000000	No
4	Germany	40.000000	63777.777778	Yes
5	France	35.000000	58000.000000	Yes
6	Spain	38.777778	52000.000000	No
7	France	48.000000	79000.000000	Yes
8	Germany	50.000000	83000.000000	No
9	France	37.000000	67000.000000	Yes

In the above line of code, it will affect the entire data-set and replaces every variable null values with their respective mean, and ‘*inplace = True*’ indicates to affect the changes to dataset.

```
In [25]: 1 dataset['Age'].fillna((dataset['Age'].mean()), inplace=True)
         2 dataset['Salary'].fillna((dataset['Salary'].mean()), inplace=True)
```

```
In [26]: 1 dataset
```

Out[26]:

	Country	Age	Salary	Purchased
0	France	44.000000	72000.000000	No
1	Spain	27.000000	48000.000000	Yes
2	Germany	30.000000	54000.000000	No
3	Spain	38.000000	61000.000000	No
4	Germany	40.000000	63777.777778	Yes
5	France	35.000000	58000.000000	Yes
6	Spain	38.777778	52000.000000	No
7	France	48.000000	79000.000000	Yes
8	Germany	50.000000	83000.000000	No
9	France	37.000000	67000.000000	Yes

If we need to replace particular variable with the strategies then we can use above line of code.

4. Divide the dataset into Dependent & Independent variable

After importing the dataset, the next step would be to identify the independent variable (X) and the dependent variable (Y).

*Basically dataset might be labeled or unlabeled, here i am considering labeled dataset for a machine learning classification problem and considering a small dataset for better understanding, in our dataset there is four columns **Country, Age, Salary and Purchased**, Actually It is a dataset of a shopping complex those handle the customer data who purchased that product or not.*

In our dataset there is three independent variables (**Country, Age and Salary**) and one dependent variable (**Purchased**) that we have to predict.

To read the columns, we will use *iloc* of *pandas* (used to fix the indexes for selection) which takes two parameters — [row selection, column selection].

```
In [6]: 1 # separate the dataset into dependent and independent variable
        2 X = dataset.iloc[:,[0,1,2]] #or X = dataset.iloc[:, :-1]
        3 Y = dataset.iloc[:,[3]] #or Y = dataset.iloc[:, -1:]
```

Note:

: selects all, using [] helps you select multiple columns or rows, this is how to slice the dataset.

You can read more about the usage of *iloc* [here](#).

This is how we were able to select the dependent variable (Y) and the independent variable (X).

5. Dealing with Categorical values

Now let's see how to deal with categorical values.

in our dataset there is one categorical variable '**Country**'. Now it gets complicated for machines to understand texts and process them, rather than numbers, since the models are based on mathematical equations and calculations. Therefore, we have to encode the categorical data.

The library that we are going to use for the task is called [Scikit Learn.preprocessing](#). There's a class in the library called *LabelEncoder* which we will use for the encoding.

```
In [34]: 1 from sklearn.preprocessing import LabelEncoder
```

The next step is usually to create an object of that class. We will call our object *lEncoder*.

```
In [35]: 1 lEncoder = LabelEncoder()
```

As you can see the first column contains data in text form. We can observe that there are 3 categories, **France, Spain & Germany**. Now to convert this into numerical we can use following code:

```
In [36]: 1 X.iloc[:,0] = lEncoder.fit_transform(X.iloc[:,0])
```

If we look at our variable X.

```
In [37]: 1 X
```

```
Out[37]:
```

	Country	Age	Salary
0	0	44.000000	72000.000000
1	2	27.000000	48000.000000
2	1	30.000000	54000.000000
3	2	38.000000	61000.000000
4	1	40.000000	63777.777778
5	0	35.000000	58000.000000
6	2	38.777778	52000.000000
7	0	48.000000	79000.000000
8	1	50.000000	83000.000000
9	0	37.000000	67000.000000

Here we can see that all three text value has been converted into numeric value:

0 : France
1 : Germany
2 : Spain

As you can see the categorical values has been encoded. **But there's a problem!**

The problem is still the same. Machine learning models are based on equations and it's good that we replaced the text by numbers.

However, since 1 is greater than 0 and 2 is greater than 1 the equations in the model will think that Spain has a higher value than Germany and France and Germany has a higher value than France. And that's certainly not the case. These are actually three categories

and there is no relational order between them. So, we have to prevent this, we're going to use **Dummy Variables**.

What are the Dummy variables?

Dummy Variables is one that takes the value 0 or 1 to indicate the absence or presence of some categorical effect that may be expected to shift the outcome.

That means instead of having one column here we are going to have three columns for each category and having 1 and 0 for their values.

Number of Columns = Types of Categories

In our case we have 3 types, so we are going to have 3 columns. To do this we will import yet another library called **OneHotEncoder**.

```
In [38]: 1 from sklearn.preprocessing import OneHotEncoder
```

Note : As OneHotEncoder requires that all values are integers, and not strings as we have. This means we first have to encode all the possible values as integers. means it should be integer values.

Next step is to create an object of that class with an important parameter called `categorical_features` which takes a value of the index of the column.

```
In [45]: 1 onehotencoder = OneHotEncoder(categorical_features=[0])
```

Using **fit_transform()** for *OneHotEncoder* as we used before for *LabelEncoder*.

```
In [48]: 1 X = onehotencoder.fit_transform(X).toarray()
```

Now we got our independent variable X in the form of numpy array. same can we check with the help of *type(X)*. just for to showing you how X look like we will converted into Pandas Dataframe with the use of below code.

```
In [53]: 1 pd.DataFrame(X, dtype=int)
```

```
Out[53]:
```

	0	1	2	3	4
0	1	0	0	44	72000
1	0	0	1	27	48000
2	0	1	0	30	54000
3	0	0	1	38	61000
4	0	1	0	40	63777
5	1	0	0	35	58000
6	0	0	1	38	52000
7	1	0	0	48	79000
8	0	1	0	50	83000
9	1	0	0	37	67000

Means there is three variables corresponding to the categories. In above code we converted all float values to integers by using *dtype*.

Country	France	Germany	Spain	Age	Salary
France	1	0	0	44	72000
Spain	0	0	1	27	48000
Germany	0	1	0	30	54000
Spain	0	0	1	38	61000
Germany	0	1	0	40	63777
France	1	0	0	35	58000
Spain	0	0	1	38	52000
France	1	0	0	48	79000
Germany	0	1	0	50	83000
France	1	0	0	37	67000

Now will doing the Label Encoding with dependent variable (Y)

```
In [68]: labelencoder_Y = LabelEncoder()
Y.iloc[:,0] = labelencoder_Y.fit_transform(Y.iloc[:,0])
```

```
In [69]: y
```

```
Out[69]:
```

Purchased	
0	0
1	1
2	0
3	0
4	1
5	1
6	0
7	1
8	0
9	1

Now we got X and Y both are in encoded form, now both can be apply on Machine Learning model.

6. Split the dataset into training and test set

In machine learning we usually splits the data into Training and Testing data for applying models.

Generally we split the dataset into 70:30 or 80:20 (as per the requirement)it means, 70 percent data taken to train and 30 percent data taken to test.

For this task, we will

import *train_test_split* from *model_selection* library of scikit.

```
In [70]: from sklearn.model_selection import train_test_split
```

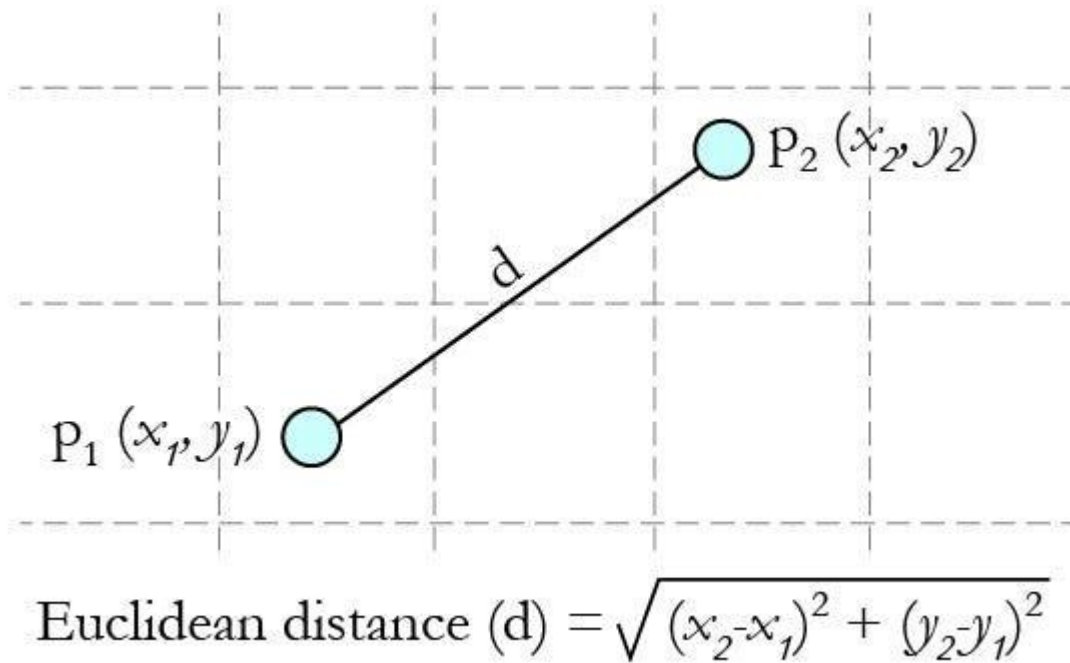
Now to build our training and test sets, we will create 4 sets — `X_train` (training part of the features), `X_test` (test part of the features), `Y_train` (training part of the dependent variables associated with the X train sets, and therefore also the same indices), `Y_test` (test part of the dependent variables associated with the X test sets, and therefore also the same indices). We will assign to them the `train_test_split`, which takes the parameters — arrays (X and Y), `test_size` (An ideal choice is to allocate 20% of the dataset to test set, it is usually assigned as 0.2. 0.25 would mean 25%).

```
In [72]: # splitting the x and y dataset into train and test set
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size = 0.25, random_state = 0)
```

7. Feature Scaling

The final step of data preprocessing is to apply the very important feature scaling.

Feature Scaling is a technique to standardize the independent features present in the data in a fixed range. It is performed during the data pre-processing.



Why Scaling :- Most of the times, your dataset will contain features highly varying in magnitudes, units and range. But since, most of the machine learning algorithms use Euclidean distance between two data points in their computations, this is a problem.

Feature Scaling is some thing which really effects the Machine Learning Model in so many ways . I agree there are so many situations where Feature Scaling is optional or not required . Still there are so many Machine Learning Algorithms where Feature Scaling is must have process . For instances — Regression, logistic regression, SVMs, k-means (see k-nearest neighbors), PCA, neural network etc.

There are so many ways to scale the feature or column value . Its completely scenario oriented that which Scalar will be more performance oriented . Lets start exploring them one by one —

Data Warehouse & Data Mining [Nitasha Rathore]

Standardization :

This is one of the most use type of scalar in data preprocessing . This is known as z-score . This re distribute the data in such a way that mean (μ) = 0 and standard deviation (σ) =1 . Here is the below formula for calculation –

$$x_{\text{stand}} = \frac{x - \text{mean}(x)}{\text{standard deviation}(x)}$$

Normalization:

Normalization scales the feature between 0.0 & 1.0, retaining their proportional range to each other.

$$x_{\text{norm}} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

The range of normal distribution is [-1,1] with mean =0.

Min-Max Scalar Technique:

Specially when you need to transform the feature magnitude in [0,1] range . This Min-Max feature scaling technique is one the best option . Here is the formula –

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

In this approach, the data is scaled to a fixed range — usually 0 to 1.

We will apply the formula of standardization and fit it to a scale. To accomplish the job, we will import the class *StandardScaler* from the *skit preprocessing* library and as usual create an object of that class.

```
In [77]: from sklearn.preprocessing import StandardScaler
```

```
In [78]: sc_x = StandardScaler()
```

Now We will transform all the data (X_train and X_test) to a same standardized scale.

```
In [82]: X_train = sc_x.fit_transform(X_train)
X_test = sc_x.fit_transform(X_test)
```

let we check how X_train look like.

```
In [81]: print(X_test)

[[ 0.          0.          0.         -1.22474487 -1.07296055]
 [ 0.          0.          0.          1.22474487  1.3343316 ]
 [ 0.          0.          0.          0.         -0.26137105]]
```

Do we need to apply feature scaling to dependent variable (Y)?

Ans: As we can see that dependent variable is categorical as it having only two value 0 and 1, and it is a classification problem so in that case we will not going to scaling this vector.

but if we will talk about for regression problem then we will do scaling with dependent variable as well.

These were the general steps for preprocessing the data. Depending on the dataset you have.

13. What is ETL?

Extract, transform, and load (ETL) is the process of combining data from multiple sources into a large, central repository called a data warehouse. ETL uses a set of business rules to clean and organize raw data and prepare it for [storage](#), [data analytics](#), and [machine learning \(ML\)](#). You can address specific business intelligence needs through data analytics (such as predicting the outcome of business decisions, generating reports and dashboards, reducing operational inefficiency, and more).

Why is ETL important?

Organizations today have both structured and unstructured data from various sources including:

- Customer data from online payment and customer relationship management (CRM) systems
- Inventory and operations data from vendor systems
- Sensor data from Internet of Things (IoT) devices
- Marketing data from social media and customer feedback
- Employee data from internal human resources systems

By applying the process of extract, transform, and load (ETL), individual raw datasets can be prepared in a format and structure that is more consumable for analytics purposes, resulting in more meaningful insights. For example, online retailers can analyze data from points of sale to forecast demand and manage inventory. Marketing teams can integrate CRM data with customer feedback on social media to study consumer behavior.

How does ETL benefit business intelligence?

Extract, transform, and load (ETL) improves business intelligence and analytics by making the process more reliable, accurate, detailed, and efficient.

Historical context

ETL gives deep historical context to the organization's data. An enterprise can combine legacy data with data from new platforms and applications. You can view older datasets alongside more recent information, which gives you a long-term view of data.

Consolidated data view

ETL provides a consolidated view of data for in-depth analysis and reporting. Managing multiple datasets demands time and coordination and can result in inefficiencies and delays. ETL combines databases and various forms of data into a single, unified view. The data integration process improves the data quality and saves the time required to move, categorize, or standardize data. This makes it easier to analyze, visualize, and make sense of large datasets.

Accurate data analysis

ETL gives more accurate data analysis to meet compliance and regulatory standards. You can integrate ETL tools with data quality tools to profile, audit, and clean data, ensuring that the data is trustworthy.

Task automation

ETL automates repeatable data processing tasks for efficient analysis. ETL tools automate the data migration process, and you can set them up to integrate data changes periodically or even at runtime. As a result, data engineers can spend more time innovating and less time managing tedious tasks like moving and formatting data.

How has ETL evolved?

Extract, transform, and load (ETL) originated with the emergence of relational databases that stored data in the form of tables for analysis. Early ETL tools attempted to convert data from transactional data formats to relational data formats for analysis.

Traditional ETL

Raw data was typically stored in transactional databases that supported many read and write requests but did not lend well to analytics. You can think of it as a row in a spreadsheet. For example, in an ecommerce system, the transactional database stored the purchased item, customer details, and order details in one transaction. Over the year, it contained a long list of transactions with repeat entries for the same customer who purchased multiple items during the year. Given the data duplication, it became cumbersome to analyze the most popular items or purchase trends in that year.

To overcome this issue, ETL tools automatically converted this transactional data into relational data with interconnected tables. Analysts could use queries to identify relationships between the tables, in addition to patterns and trends.

Modern ETL

As ETL technology evolved, both data types and data sources increased exponentially. Cloud technology emerged to create vast databases (also called data sinks). Such data sinks can receive data from multiple sources and have underlying hardware resources that can scale over time. ETL tools have also become more sophisticated and can work with modern data sinks. They can convert data from legacy data formats to modern data formats. Examples of modern databases follow.

Data warehouses

A [data warehouse](#) is a central repository that can store multiple databases. Within each database, you can organize your data into tables and columns that describe the data types in the table. The data warehouse software works across multiple types of storage hardware—such as solid state drives (SSDs), hard drives, and other cloud storage—to optimize your data processing.

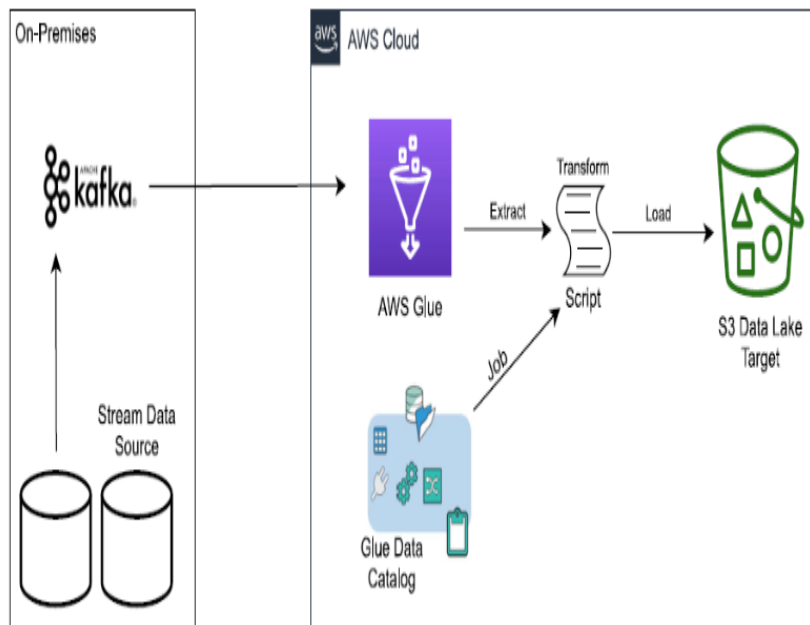
Data lakes

With a [data lake](#), you can store your structured and unstructured data in one centralized repository and at any scale. You can store data as is without having to first structure it based on questions you might have in the future. Data lakes also allow you to run different types of analytics on your data, like SQL queries, big data analytics, full-text search, real-time analytics, and machine learning (ML) to guide better decisions.

How does ETL work?

Extract, transform, and load (ETL) works by moving data from the source system to the destination system at periodic intervals. The ETL process works in three steps:

1. Extract the relevant data from the source database
2. Transform the data so that it is better suited for analytics
3. Load the data into the target database



What is data extraction?

In data extraction, extract, transform, and load (ETL) tools extract or copy raw data from multiple sources and store it in a staging area. A staging area (or landing zone) is an intermediate storage area for temporarily storing extracted data. Data staging areas are often transient, meaning their contents are erased after data extraction is complete. However, the staging area might also retain a data archive for troubleshooting purposes.

How frequently the system sends data from the data source to the target data store depends on the underlying change data capture mechanism. Data extraction commonly happens in one of the three following ways.

Update notification

In update notification, the source system notifies you when a data record changes. You can then run the extraction process for that change. Most databases and web applications provide update mechanisms to support this data integration method.

Incremental extraction

Some data sources can't provide update notifications but can identify and extract data that has been modified over a given time period. In this case, the system checks for changes at periodic intervals, such as once a week, once a month, or at the end of a campaign. You only need to extract data that has changed.

Full extraction

Some systems can't identify data changes or give notifications, so reloading all data is the only option. This extraction method requires you to keep a copy of the last extract to check which records are new. Because this approach involves high data transfer volumes, we recommend you use it only for small tables.

What is data transformation?

In data transformation, extract, transform, and load (ETL) tools transform and consolidate the raw data in the staging area to prepare it for the target data warehouse. The data transformation phase can involve the following types of data changes.

Basic data transformation

Basic transformations improve data quality by removing errors, emptying data fields, or simplifying data. Examples of these transformations follow.

Data cleansing

Data cleansing removes errors and maps source data to the target data format. For example, you can map empty data fields to the number 0, map the data value "Parent" to "P," or map "Child" to "C."

Data deduplication

Deduplication in data cleansing identifies and removes duplicate records.

Data format revision

Format revision converts data, such as character sets, measurement units, and date/time values, into a consistent format. For example, a food company might have different recipe databases with ingredients measured in kilograms and pounds. ETL will convert everything to pounds.

Advanced data transformation

Advanced transformations use business rules to optimize the data for easier analysis. Examples of these transformations follow.

Derivation

Derivation applies business rules to your data to calculate new values from existing values. For example, you can convert revenue to profit by subtracting expenses or calculating the total cost of a purchase by multiplying the price of each item by the number of items ordered.

Joining

In data preparation, joining links the same data from different data sources. For example, you can find the total purchase cost of one item by adding the purchase value from different vendors and storing only the final total in the target system.

Splitting

You can divide a column or data attribute into multiple columns in the target system. For example, if the data source saves the customer name as “Jane John Doe,” you can split it into a first, middle, and last name.

Summarization

Summarization improves data quality by reducing a large number of data values into a smaller dataset. For example, customer order invoice values can have many different small amounts. You can summarize the data by adding them up over a given period to build a customer lifetime value (CLV) metric.

Encryption

You can protect sensitive data to comply with data laws or data privacy by adding encryption before the data streams to the target database.

What is data loading?

In data loading, extract transform, and load (ETL) tools move the transformed data from the staging area into the target data warehouse. For most organizations that use ETL, the process is automated, well defined, continual, and batch driven. Two methods for loading data follow.

Full load

In full load, the entire data from the source is transformed and moved to the data warehouse. The full load usually takes place the first time you load data from a source system into the data warehouse.

Incremental load

In incremental load, the ETL tool loads the delta (or difference) between target and source systems at regular intervals. It stores the last extract date so that only records added after this date are loaded. There are two ways to implement incremental load.

Streaming incremental load

If you have small data volumes, you can stream continual changes over data pipelines to the target data warehouse. When the speed of data increases to millions of events per second, you can use event stream processing to monitor and process the data streams to make more-timely decisions.

Batch incremental load

If you have large data volumes, you can collect load data changes into batches periodically. During this set period of time, no actions can happen to either the source or target system as data is synchronized.

What is ELT?

Extract, load, and transform (ELT) is an extension of extract, transform, and load (ETL) that reverses the order of operations. You can load data directly into the target system before processing it. The intermediate staging area is not required because the target data warehouse

has data mapping capabilities within it. ETL has become more popular with the adoption of cloud infrastructure, which gives target databases the processing power they need for transformations.

ETL compared to ELT

ELT works well for high-volume, unstructured datasets that require frequent loading. It is also ideal for big data because the planning for analytics can be done after data extraction and storage. It leaves the bulk of transformations for the analytics stage and focuses on loading minimally processed raw data into the data warehouse.

The ETL process requires more definition at the beginning. Analytics needs to be involved from the start to define target data types, structures, and relationships. Data scientists mainly use ETL to load legacy databases into the warehouse, and ELT has become the norm today.

What is data virtualization?

Data virtualization uses a software abstraction layer to create an integrated data view without physically extracting, transforming, or loading the data. Organizations use this functionality as a virtual unified data repository without the expense and complexity of building and managing separate platforms for source and target. While you can use data virtualization alongside extract, transform, and load (ETL), it is increasingly seen as an alternative to ETL and other physical data integration methods. For example, you can use [AWS Glue Elastic Views](#) to quickly create a virtual table—a materialized view—from multiple different source data stores.

14. Dimensional Data Modeling

- **Dimensional Data Modeling** is one of the data modeling techniques used in data warehouse design. The concept of Dimensional Modeling was developed by Ralph Kimball which is comprised of facts and dimension tables. Since the main goal of this modeling is to improve the data retrieval so it is optimized for SELECT OPERATION. The advantage of using this model is that we can store data in such a way that it is easier to store and retrieve the data once stored in a data warehouse. The dimensional model is the data model used by many OLAP systems.

Elements of Dimensional Data Model

Facts

Facts are the measurable data elements that represent the business metrics of interest. For example, in a sales data warehouse, the facts might include sales revenue, units sold, and profit margins. Each fact is associated with one or more dimensions, creating a relationship between the fact and the descriptive data.

Dimension

Dimensions are the descriptive data elements that are used to categorize or classify the data. For example, in a sales data warehouse, the dimensions might include product, customer, time, and location. Each dimension is made up of a set of

attributes that describe the dimension. For example, the product dimension might include attributes such as product name, product category, and product price.

Attributes

Characteristics of dimension in data modeling are known as characteristics. These are used to filter, search facts, etc. For a dimension of location, attributes can be State, Country, Zipcode, etc.

Fact Table

In a dimensional data model, the fact table is the central table that contains the measures or metrics of interest, surrounded by the dimension tables that describe the attributes of the measures. The dimension tables are related to the fact table through foreign key relationships

Dimension Table

Dimensions of a fact are mentioned by the dimension table and they are basically joined by a [foreign key](#). Dimension tables are simply de-normalized tables. The dimensions can be having one or more relationships.

Types of Dimensions in Data Warehouse Model

- Conformed Dimension
- Outtrigger Dimension
- Shrunk Dimension
- Role-Playing Dimension
- Dimension to Dimension Table
- Junk Dimension
- Degenerate Dimension
- Swappable Dimension
- Step Dimension

Steps to Create Dimensional Data Modeling

Step-1: Identifying the business objective: The first step is to identify the business objective. Sales, HR, Marketing, etc. are some examples of the need of the organization. Since it is the most important step of Data Modelling the selection of business objectives also depends on the quality of data available for that process.

Step-2: Identifying Granularity: Granularity is the lowest level of information stored in the table. The level of detail for business problems and its solution is described by Grain.

Step-3: Identifying Dimensions and their Attributes: Dimensions are objects or things. Dimensions categorize and describe data warehouse facts and measures in a way that supports meaningful answers to business questions. A data warehouse organizes descriptive attributes as columns in dimension tables. For Example, the data dimension may contain data like a year, month, and weekday.

Step-4: Identifying the Fact: The measurable data is held by the fact table. Most of the fact table rows are numerical values like price or cost per unit, etc.

Step-5: Building of Schema: We implement the Dimension Model in this step. A schema is a database structure. There are two popular schemes: [Star Schema](#) and [Snowflake Schema](#).

Dimensional Data Modeling Steps

Dimensional data modeling is a technique used in data warehousing to organize and structure data in a way that makes it easy to analyze and understand. In a dimensional data model, data is organized into dimensions and facts.

Overall, dimensional data modeling is an effective technique for organizing and structuring data in a data warehouse for analysis and reporting. By providing a simple and intuitive structure for the data, the dimensional model makes it easy for users to access and understand the data they need to make informed business decisions

Advantages of Dimensional Data Modeling

- **Simplified Data Access:** Dimensional data modeling enables users to easily access data through simple queries, reducing the time and effort required to retrieve and analyze data.
- **Enhanced Query Performance:** The simple structure of dimensional data modeling allows for faster query performance, particularly when compared to relational data models.
- **Increased Flexibility:** Dimensional data modeling allows for more flexible data analysis, as users can quickly and easily explore relationships between data.
- **Improved Data Quality:** Dimensional data modeling can improve data quality by reducing redundancy and inconsistencies in the data.
- **Easy to Understand:** Dimensional data modeling uses simple, intuitive structures that are easy to understand, even for non-technical users.

Disadvantages of Dimensional Data Modeling

- **Limited Complexity:** Dimensional data modeling may not be suitable for very complex data relationships, as it relies on simple structures to organize data.
- **Limited Integration:** Dimensional data modeling may not integrate well with other data models, particularly those that rely on normalization techniques.
- **Limited Scalability:** Dimensional data modeling may not be as scalable as other data modeling techniques, particularly for very large datasets.
- **Limited History Tracking:** Dimensional data modeling may not be able to track changes to historical data, as it typically focuses on current data.

15. **REQUIREMENTS TO DATA DESIGN**

The requirements definition completely drives the data design for the data warehouse. Data design consists of putting together the data structures. A group of data elements form a data structure. Logical data design includes determination of the various data elements that are needed and combination of the data elements into structures of data. Logical data design also includes establishing the relationships among the data structures. Let us look at Figure 10-1. Notice how the phases start with requirements gathering. The results of the requirements gathering phase is documented in detail in the requirements definition document. An essential component of this document is the set of information package diagrams. Remember that these are information matrices showing the metrics, business dimensions, and the hierarchies within individual business dimensions. The information package diagrams form the basis for

the logical data design for the data warehouse. The data design process results in a dimensional data model.

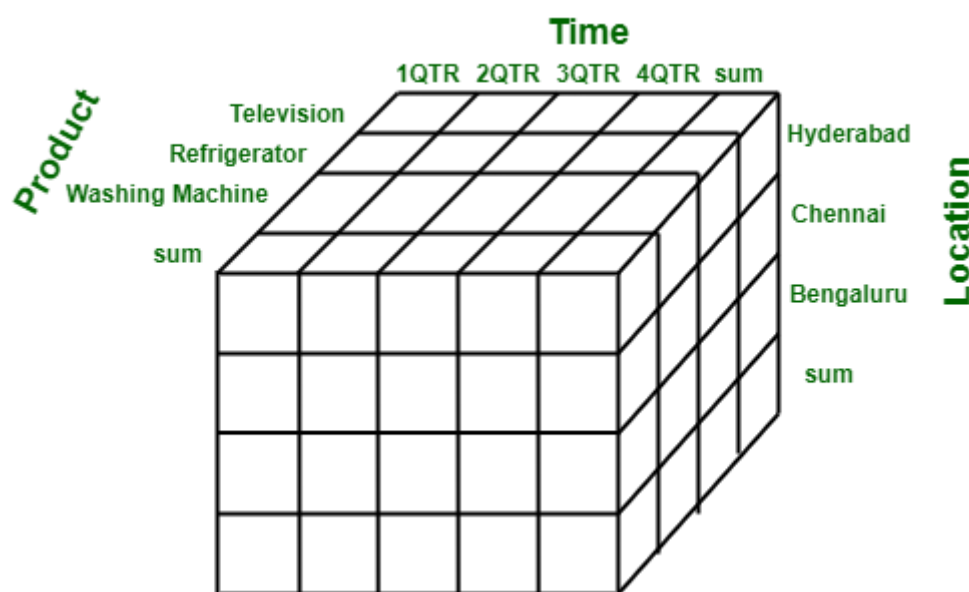
16. MultiDimensional Data Model

The multi-Dimensional Data Model is a method which is used for ordering data in the database along with good arrangement and assembling of the contents in the database.

The Multi-Dimensional Data Model allows customers to interrogate analytical questions associated with market or business trends, unlike relational databases which allow customers to access data in the form of queries. They allow users to rapidly receive answers to the requests which they made by creating and examining the data comparatively fast.

OLAP (online analytical processing) and data warehousing uses multi dimensional databases. It is used to show multiple dimensions of the data to users.

It represents data in the form of data cubes. Data cubes allow to model and view the data from many dimensions and perspectives. It is defined by dimensions and facts and is represented by a fact table. Facts are numerical measures and fact tables contain measures of the related dimensional tables or names of the facts.



Multidimensional Data Representation

Working on a Multidimensional Data Model

On the basis of the pre-decided steps, the Multidimensional Data Model works.

The following stages should be followed by every project for building a Multi Dimensional Data Model :

Stage 1 : Assembling data from the client : In first stage, a Multi Dimensional Data Model collects correct data from the client. Mostly, software professionals provide simplicity to the client about the range of data which can be gained with the selected technology and collect the complete data in detail.

Stage 2 : Grouping different segments of the system : In the second stage, the Multi Dimensional Data Model recognizes and classifies all the data to the respective section they belong to and also builds it problem-free to apply step by step.

Stage 3 : Noticing the different proportions : In the third stage, it is the basis on which the design of the system is based. In this stage, the main factors are recognized according to the user's point of view. These factors are also known as "Dimensions".

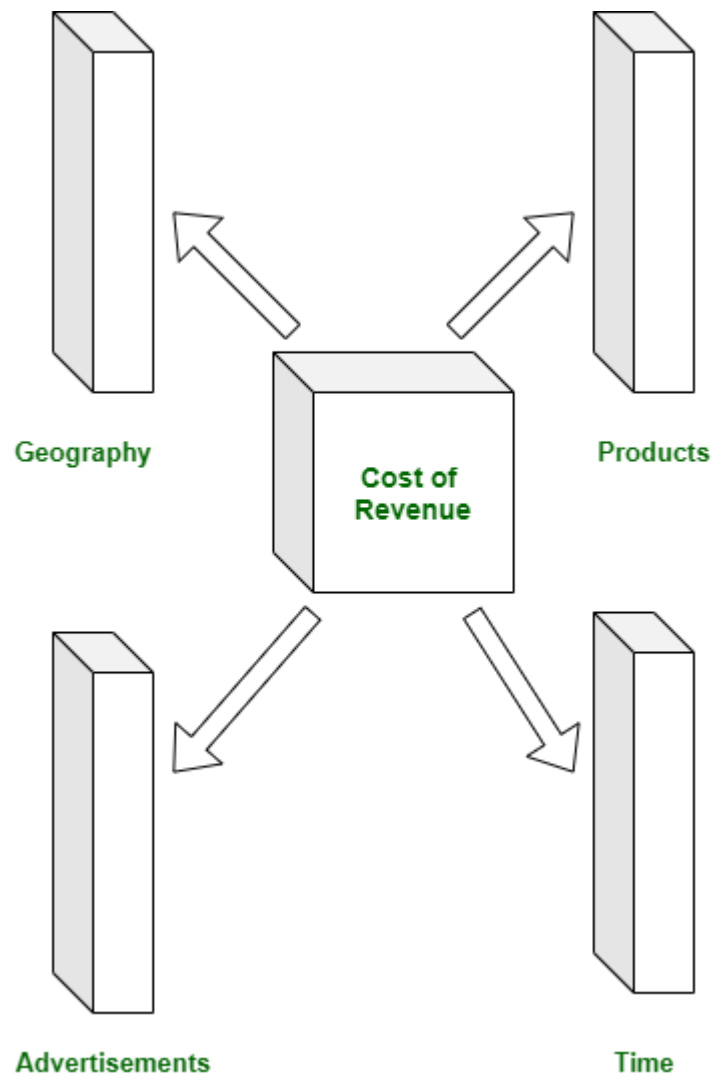
Stage 4 : Preparing the actual-time factors and their respective qualities : In the fourth stage, the factors which are recognized in the previous step are used further for identifying the related qualities. These qualities are also known as "**attributes**" in the database.

Stage 5 : Finding the actuality of factors which are listed previously and their qualities : In the fifth stage, A Multi Dimensional Data Model separates and differentiates the actuality from the factors which are collected by it. These actually play a significant role in the arrangement of a Multi Dimensional Data Model.

Stage 6 : Building the Schema to place the data, with respect to the information collected from the steps above : In the sixth stage, on the basis of the data which was collected previously, a Schema is built.

For Example :

1. Let us take the example of a firm. The revenue cost of a firm can be recognized on the basis of different factors such as geographical location of firm's workplace, products of the firm, advertisements done, time utilized to flourish a product, etc.



Example 1

2. Let us take the example of the data of a factory which sells products per quarter in Bangalore. The data is represented in the table given below :

Location = "Bangalore"				
Time (quarter)	Type of item			
	Jam	Bread	Sugar	Milk
Q1	350	389	35	50
Q2	260	528	50	90
Q3	483	256	20	60
Q4	436	396	15	40

2D factory data

In the above given presentation, the factory's sales for Bangalore are, for the time dimension, which is organized into quarters and the dimension of

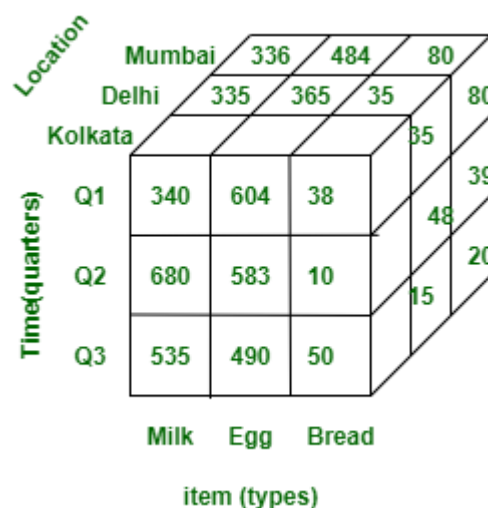
items, which is sorted according to the kind of item which is sold. The facts here are represented in rupees (in thousands).

Now, if we desire to view the data of the sales in a three-dimensional **table**, then it is represented in the diagram given below. Here the data of the sales is represented as a two **dimensional table**. Let us consider the data according to item, time and location (like Kolkata, Delhi, Mumbai). Here is the table :

Time	Location="Kolkata"			Location="Delhi"			Location="Mumbai"		
	item			item			item		
	Milk	Egg	Bread	Milk	Egg	Bread	Milk	Egg	Bread
Q1	340	604	38	335	365	35	336	484	80
Q2	680	583	10	684	490	48	595	594	39
Q3	535	490	50	389	385	15	366	385	20

3D data representation as 2D

This data can be represented in the form of three dimensions conceptually, which is shown in the image below :



3D data representation

Features of multidimensional data models:

Measures: Measures are numerical data that can be analyzed and compared, such as sales or revenue. They are typically stored in fact tables in a multidimensional data model.

Dimensions: Dimensions are attributes that describe the measures, such as time, location, or product. They are typically stored in dimension tables in a multidimensional data model.

Cubes: Cubes are structures that represent the multidimensional relationships between measures and dimensions in a data model. They provide a fast and efficient way to retrieve and analyze data.

Aggregation: Aggregation is the process of summarizing data across dimensions and levels of detail. This is a key feature of multidimensional data models, as it enables users to quickly analyze data at different levels of granularity.

Drill-down and roll-up: Drill-down is the process of moving from a higher-level summary of data to a lower level of detail, while roll-up is the opposite process of moving from a lower-level detail to a higher-level summary. These features enable users to explore data in greater detail and gain insights into the underlying patterns.

Hierarchies: Hierarchies are a way of organizing dimensions into levels of detail. For example, a time dimension might be organized into years, quarters, months, and days. Hierarchies provide a way to navigate the data and perform drill-down and roll-up operations.

OLAP (Online Analytical Processing): OLAP is a type of multidimensional data model that supports fast and efficient querying of large datasets. OLAP systems are designed to handle complex queries and provide fast response times.

Advantages of Multi Dimensional Data Model

The following are the advantages of a multi-dimensional data model :

- A multi-dimensional data model is easy to handle.
- It is easy to maintain.
- Its performance is better than that of normal databases (e.g. relational databases).
- The representation of data is better than traditional databases. That is because the multi-dimensional databases are multi-viewed and carry different types of factors.
- It is workable on complex systems and applications, contrary to the simple one-dimensional database systems.
- The compatibility in this type of database is an upliftment for projects having lower bandwidth for maintenance staff.

Disadvantages of Multi Dimensional Data Model

The following are the disadvantages of a Multi Dimensional Data Model :

- The multi-dimensional Data Model is slightly complicated in nature and it requires professionals to recognize and examine the data in the database.
- During the work of a Multi-Dimensional Data Model, when the system caches, there is a great effect on the working of the system.
- It is complicated in nature due to which the databases are generally dynamic in design.
- The path to achieving the end product is complicated most of the time.
- As the Multi Dimensional Data Model has complicated systems, databases have a large number of databases due to which the system is very insecure when there is a security break.

17. Star schema

Star schema is one of three schemas used in data warehouses to properly organize your data. The other two are snowflake schema and galaxy schema. While there are differences between the three, they all share a similarity in that they all promote fast and efficient querying of large data sets.

The descriptions below outline how each of these schemas functions in practice and explore them with a simple example of credit card payments data from Burst Bank.

Star schema example

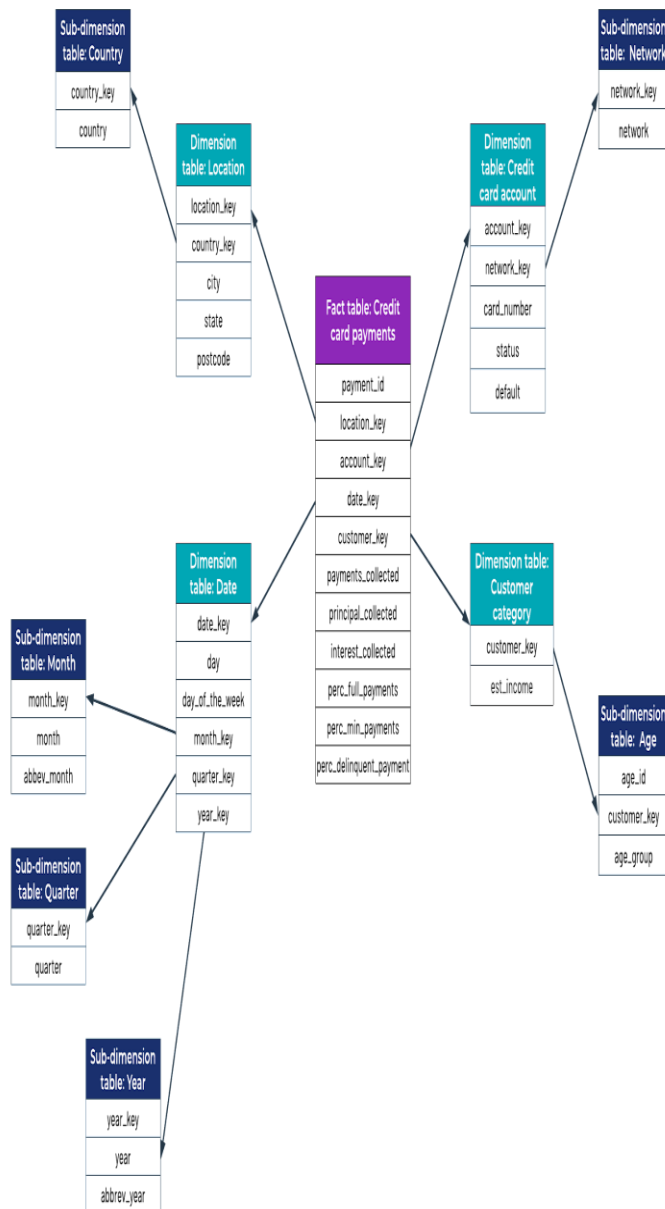
In the figure below, the credit card payments fact table is related to four dimensions: location, date, credit card account, and customer group. Customer group includes estimated income and age. The fact table also includes many aggregate measures of data credit card payment data, such as the percent of full payments collected and the percent of minimum payments collected. Business analysts can use these measures to answer the types of questions that will help their company make business decisions. An example of a question that they could answer with this data is: Are customers over the age of 40 more likely than customers under 40 to make full payments?

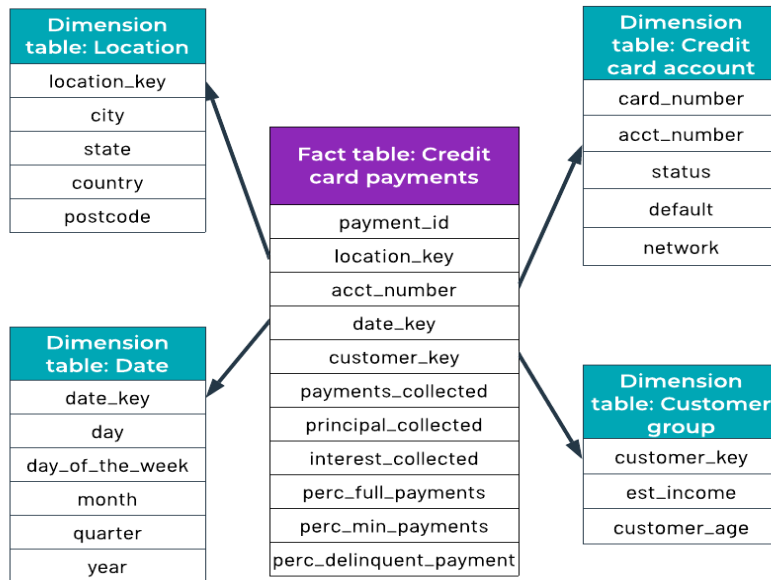
18. Snowflake schema

In a snowflake schema, each dimension connects to a central fact table. Some of the dimension tables are normalized so that they have sub-dimensions. This normalization reduces some of the data redundancy and makes it easier to maintain and store data. However, this comes at the cost of less efficient queries because each query requires more tables to be joined together.

Snowflake schema example

In the figure below, the credit card payments fact table is related to the same dimensions as in the star schema example. However, each dimension table is also connected to one or more sub-dimension tables. For example, the date dimension table has sub-dimensions for month, quarter, and year. The sub-dimension tables for month and year include abbreviations that business analysts might use to look up month or year from the database.

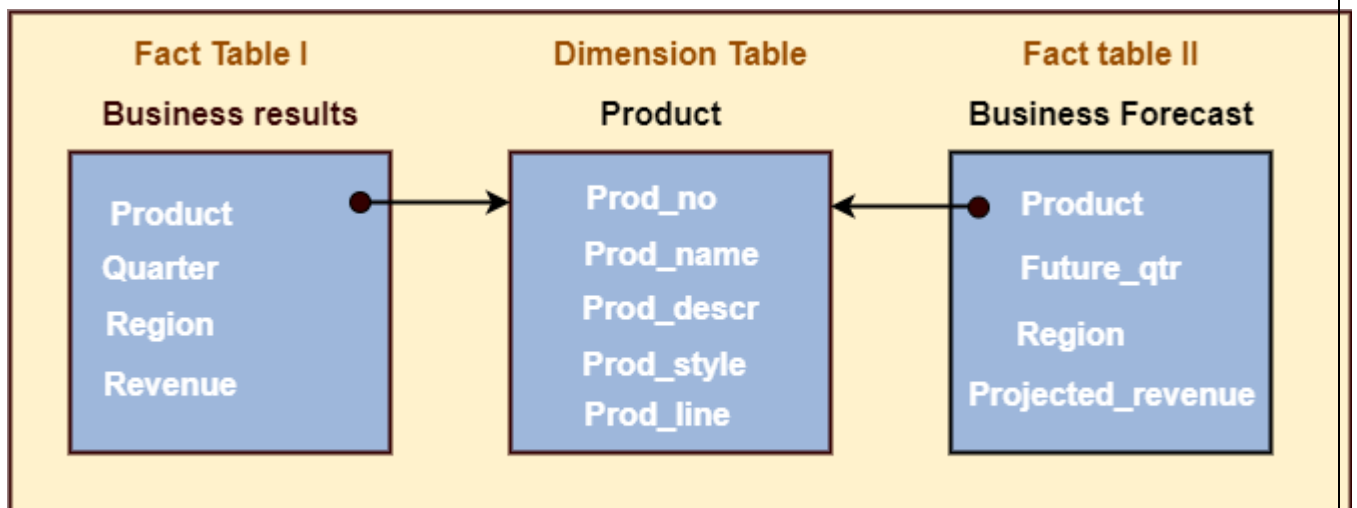




19. Fact Constellation Schema

A Fact constellation means two or more fact tables sharing one or more dimensions. It is also called **Galaxy schema**.

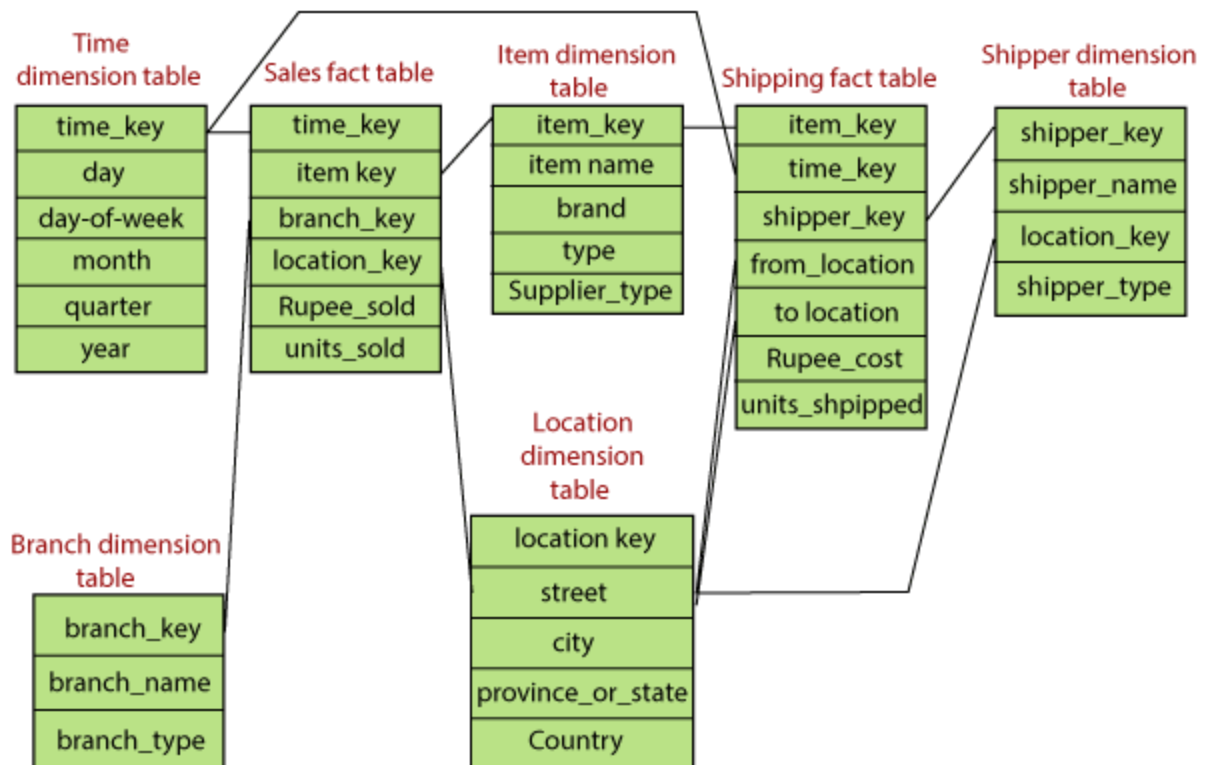
Fact Constellation Schema describes a logical structure of data warehouse or data mart. Fact Constellation Schema can design with a collection of de-normalized FACT, Shared, and Conformed Dimension tables.



FACT Constellation Schema

Fact Constellation Schema is a sophisticated database design that is difficult to summarize information. Fact Constellation Schema can implement between aggregate Fact tables or decompose a complex Fact table into independent simplex Fact tables.

Example: A fact constellation schema is shown in the figure below.



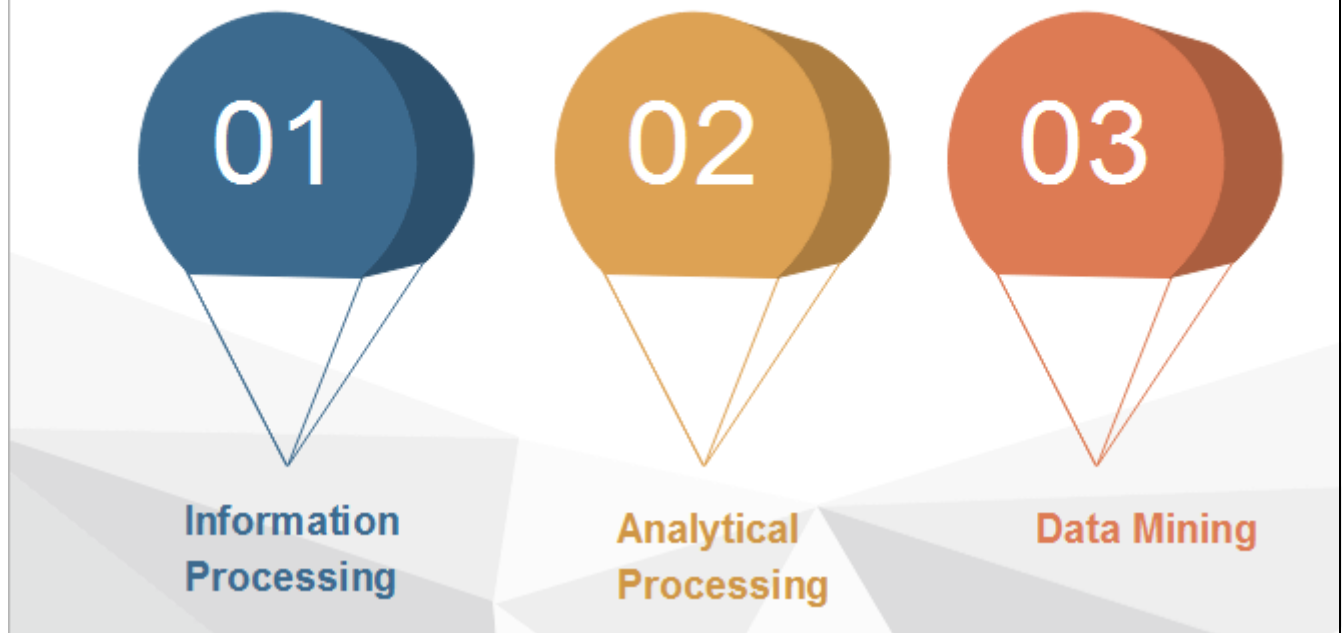
This schema defines two fact tables, sales, and shipping. Sales are treated along four dimensions, namely, time, item, branch, and location. The schema contains a fact table for sales that includes keys to each of the four dimensions, along with two measures: Rupee_sold and units_sold. The shipping table has five dimensions, or keys: item_key, time_key, shipper_key, from_location, and to_location, and two measures: Rupee_cost and units_shipped.

The primary disadvantage of the fact constellation schema is that it is a more challenging design because many variants for specific kinds of aggregation must be considered and selected.

Data Warehouse Applications

The application areas of the data warehouse are:

Data Warehouse Applications



Information Processing

It deals with querying, statistical analysis, and reporting via tables, charts, or graphs. Nowadays, information processing of data warehouse is to construct a low cost, web-based accessing tools typically integrated with web browsers.

Analytical Processing

It supports various online analytical processing such as drill-down, roll-up, and pivoting. The historical data is being processed in both summarized and detailed format.

OLAP is implemented on data warehouses or data marts. The primary objective of OLAP is to support ad-hoc querying needed for support DSS. The multidimensional view of data is fundamental to the OLAP application. OLAP is an operational view, not a data structure or schema. The complex nature of OLAP applications requires a multidimensional view of the data.

Data Mining

It helps in the analysis of hidden design and association, constructing scientific models, operating classification and prediction, and performing the mining results using visualization tools.

Data mining is the technique of designing essential new correlations, patterns, and trends by changing through high amounts of a record save in repositories, using pattern recognition technologies as well as statistical and mathematical techniques.

It is the phase of selection, exploration, and modeling of huge quantities of information to determine regularities or relations that are at first unknown to access precise and useful results for the owner of the database.

It is the process of inspection and analysis, by automatic or semi-automatic means, of large quantities of records to discover meaningful patterns and rules

32. Data Warehouse and Business Strategy Data warehousing (DW) is a method of gathering and analyzing data from many sources in order to get useful business insights. Typically, a data warehouse is used to integrate and analyze corporate data from many sources. The data warehouse is the heart of the business intelligence (BI) system, which is designed to analyze and report on data.

It's a collection of technology and components that helps with data strategy. It refers to a company's electronic storage of a huge volume of data that is intended for inquiry and analysis rather than transaction processing. It is a method of converting data into information and making it available to people in a timely manner so that it can be used to make a difference.

By analyzing a set of data in a structured manner, you can extract more accurate information for use in determining a company's stance on a certain issue. It's for this reason that gudang data is one of the most important parameters in business intelligence.

How Does a Data Warehouse Work?

A Data Warehouse is a central repository where data from one or more data sources is stored. The transactional system and other relational databases feed data into a data warehouse. The examples of data sources, such as Structured Data, Semi-Structured Data, and Unstructured Data.

Users can access the processed data in the Data Warehouse using Business Intelligence tools, SQL clients, and spreadsheets after the data has been processed, transformed, and ingested. A data warehouse combines data from various sources into a single, comprehensive database.

An organization may examine its consumers more holistically by combining all of this information in one place. This ensures that it has taken into account all of the accessible data. Data mining is made possible by data warehousing. Data mining is the process of looking for patterns in data that could lead to new insights.

Types of Data Warehouse



The following are the three primary types of Data Warehouses (DWH):

- **Enterprise Data Warehouse (EDW):** An EDW is a data warehouse that stores information from multiple sources. A centralized warehouse is an Enterprise Data Warehouse (EDW). It offers decision-making assistance to the entire company. It provides a uniform approach to data organization and representation. It also has the capability of classifying data by subject and granting access based on such divisions.
- **Data Store for Operational Purposes:** Operational Data Stores, or ODS, are simply data stores that are required when neither a data warehouse nor an OLTP system can meet an organization's reporting needs. The data warehouse in ODS is updated in real time. As a result, it is extensively used for mundane tasks such as keeping employee details.
- **Data Mart:** The data warehouse is subdivided into data marts. It is tailored to a specific business line, such as sales, finance, sales, or finance. Data can be collected directly from sources in an independent data mart.

The Benefit of Using Data Warehouse



Improved data analytics, more revenue, and the capacity to compete more strategically in the marketplace are all advantages of a data warehouse. A data warehouse drives a more effective data strategy by efficiently supplying standardized, contextual data to an organization's business intelligence software. The following are some of the benefits that can be obtained by implementing a data warehouse.

- **Data Analytics Can Be Made More Powerful And Faster**

Business intelligence and data analytics are the reverse of instinct and intuition. High-quality, consistent data is required for BI and analytics, and it must be delivered on time and be available for data mining quickly. This strength and speed are enabled by a data warehouse, which provides a competitive advantage in important business sectors such as CRM, HR, sales success, and quarterly reporting.

- **Increases Revenue**

Data Warehouse & Data Mining [Nitasha Rathore]

A data warehouse's primary strength is creating more standardized and higher-quality data, and this critical strength correlates to large income advantages. The data warehouse formula is as follows: better business intelligence leads to better judgments, and better decisions lead to a higher return on investment in every sector of your company.

Most importantly, these income increases compound over time as the firm improves as a result of better decisions. In other words, a high-quality, fully scalable data warehouse is more of an investment than an expense – one that adds exponential value like few others.

- **Interacts with both on-premise and cloud-based systems**

Today's data warehouses, unlike traditional databases of the past, are designed with multi cloud and hybrid cloud in mind. Many data warehouses are now completely cloud-based, and even those that were designed for on-premise will often work well with a company's cloud-based infrastructure. As a side note, this cloud-based approach also implies that mobile users will have easier access to the data warehouse, which will benefit sales reps in particular.

- **Enhances Data Consistency And Quality**

Usually a business creates data in a variety of formats, including structured and unstructured data, social media data, and sales campaign data. A data warehouse translates this information into the formats that your analytics tools require. Furthermore, a data warehouse ensures that data provided by multiple business divisions is of the same quality and standard, allowing for a more efficient analytics feed.

- **Enhances Productivity**

Gathering data from numerous sources takes a long time for a business user or a data scientist. It's significantly more convenient to have all of this information in one location, which is why a data warehouse is so useful.

Furthermore, if your data scientist requires data to run a quick report, they do not require assistance from tech support to complete this activity. A data warehouse makes this information easily available — in the correct format – boosting the overall efficiency of the process.

- **Query Performance And Insights Have Significantly Improved**

Constant business intelligence queries, from traditional databases to data marts, can impose a burden on an analytics infrastructure. The usage of a data warehouse to better handle queries relieves some of the strain on the system.

Furthermore, because a data warehouse is designed to handle large amounts of data and a variety of complex queries, it is the backbone of every company's data analytics strategy.

- **Data Protection**

A number of significant advancements in data warehouse security have improved the overall security of enterprise data. Techniques like a "slave read only" setup, which prohibits dangerous SQL code, and encrypted columns, which secure personal data, are examples of these advancements.

On their data warehouses, some companies create unique user groups that can include or exclude specific data pools and even grant permission row by row.

- **Offers a Significant Competitive Advantage**

The bottom line benefit of a data warehouse is that it allows a company to strategize and execute more effectively against competitors in its industry. Greater insight in data mining can drive decisions that result in higher sales, better focused products, and faster response times, thanks to the quality, speed, and historical context supplied by a data warehouse. In short, a data warehouse helps businesses make better decisions, which offers them a competitive advantage.

20.What is online analytical processing?

Online analytical processing (OLAP) is software technology you can use to analyze business data from different points of view. Organizations collect and store data from multiple data sources, such as websites, applications, smart meters, and internal systems. OLAP combines and groups this data into categories to provide actionable insights for strategic planning. For example, a retailer stores data about all the products it sells, such as color, size, cost, and location. The retailer also collects customer purchase data, such as the name of the items ordered and total sales value, in a different system. OLAP combines the datasets to answer questions such as which color products are more popular or how product placement impacts sales.

Why is OLAP important?

Online analytical processing (OLAP) helps organizations process and benefit from a growing amount of digital information. Some benefits of OLAP include the following.

Faster decision making

Businesses use OLAP to make quick and accurate decisions to remain competitive in a fast-paced economy. Performing analytical queries on multiple relational databases is time consuming because the computer system searches through multiple data tables. On the other hand, OLAP systems precalculate and integrate data so business analysts can generate reports faster when needed.

Non-technical user support

OLAP systems make complex data analysis easier for non-technical business users. Business users can create complex analytical calculations and generate reports instead of learning how to operate databases.

Integrated data view

OLAP provides a unified platform for marketing, finance, production, and other business units. Managers and decision makers can see the bigger picture and effectively solve problems. They can perform what-if analysis, which shows the impact of decisions taken by one department on other areas of the business.

What is OLAP architecture?

Online analytical processing (OLAP) systems store multidimensional data by representing information in more than two dimensions, or categories. Two-dimensional data involves columns and rows, but multidimensional data has multiple characteristics. For example, multidimensional data for product sales might consist of the following dimensions:

- Product type
- Location
- Time

Data engineers build a multidimensional OLAP system that consists of the following elements.

Data warehouse

A data warehouse collects information from different sources, including applications, files, and databases. It processes the information using various tools so that the data is ready for analytical purposes. For example, the data warehouse might collect information from a relational database that stores data in tables of rows and columns.

ETL tools

Extract, transform, and load (ETL) tools are database processes that automatically retrieve, change, and prepare the data to a format fit for analytical purposes. Data warehouses use ETL to convert and standardize information from various sources before making it available to OLAP tools.

OLAP server

An OLAP server is the underlying machine that powers the OLAP system. It uses ETL tools to transform information in the relational databases and prepare them for OLAP operations.

OLAP database

An OLAP database is a separate database that connects to the data warehouse. Data engineers sometimes use an OLAP database to prevent the data warehouse from being burdened by OLAP analysis. They also use an OLAP database to make it easier to create OLAP data models.

OLAP cubes

A data cube is a model representing a multidimensional array of information. While it's easier to visualize it as a three-dimensional data model, most data cubes have more than three dimensions. An OLAP cube, or hypercube, is the term for data cubes in an OLAP system. OLAP cubes are rigid because you can't change the dimensions and underlying data once you model it. For example, if you add the warehouse dimension to a cube with product, location, and time dimensions, you have to remodel the entire cube.

OLAP analytic tools

Business analysts use OLAP tools to interact with the OLAP cube. They perform operations such as slicing, dicing, and pivoting to gain deeper insights into specific information within the OLAP cube.

How does OLAP work?

How does OLAP work?

An online analytical processing (OLAP) system works by collecting, organizing, aggregating, and analyzing data using the following steps:

1. The OLAP server collects data from multiple data sources, including relational databases and data warehouses.
2. Then, the extract, transform, and load (ETL) tools clean, aggregate, precalculate, and store data in an OLAP cube according to the number of dimensions specified.
3. Business analysts use OLAP tools to query and generate reports from the multidimensional data in the OLAP cube.

OLAP uses Multidimensional Expressions (MDX) to query the OLAP cube. MDX is a query, like SQL, that provides a set of instructions for manipulating databases.

What are the types of OLAP?

Online analytical processing (OLAP) systems operate in three main ways.

MOLAP

Multidimensional online analytical processing (MOLAP) involves creating a data cube that represents multidimensional data from a data warehouse. The MOLAP system stores precalculated data in the hypercube. Data engineers use MOLAP because this type of OLAP technology provides fast analysis.

ROLAP

Instead of using a data cube, relational online analytical processing (ROLAP) allows data engineers to perform multidimensional data analysis on a relational database. In other words, data engineers use SQL queries to search for and retrieve specific information based on the required dimensions. ROLAP is suitable for analyzing extensive and detailed data. However, ROLAP has slow query performance compared to MOLAP.

Data Warehouse & Data Mining [Nitasha Rathore]

HOLAP

Hybrid online analytical processing (HOLAP) combines MOLAP and ROLAP to provide the best of both architectures. HOLAP allows data engineers to quickly retrieve analytical results from a data cube and extract detailed information from relational databases.

What is data modeling in OLAP?

Data modeling is the representation of data in data warehouses or online analytical processing (OLAP) databases. Data modeling is essential in relational online analytical processing (ROLAP) because it analyzes data straight from the relational database. It stores multidimensional data as a star or snowflake schema.

Star schema

The star schema consists of a fact table and multiple dimension tables. The fact table is a data table that contains numerical values related to a business process, and the dimension table contains values that describe each attribute in the fact table. The fact table refers to dimensional tables with foreign keys—unique identifiers that correlate to the respective information in the dimension table.

In a star schema, a fact table connects to several dimension tables so the data model looks like a star. The following is an example of a fact table for product sales:

- Product ID
- Location ID
- Salesperson ID
- Sales amount

The product ID tells the database system to retrieve information from the product dimension table, which might look as follows:

- Product ID
- Product name
- Product type
- Product cost

Likewise, the location ID points to a location dimension table, which could consist of the following:

- Location ID
- Country
- City

The salesperson table might look as follows:

- Salesperson ID
- First name
- Last name
- Email

Snowflake schema

The snowflake schema is an extension of the star schema. Some dimension tables might lead to one or more secondary dimension tables. This results in a snowflake-like shape when the dimension tables are put together.

For example, the product dimension table might contain the following fields:

- Product ID
- Product name
- Product type ID
- Product cost

The product type ID connects to another dimension table as shown in the following example:

- Product type ID
- Type name
- Version
- Variant

What are OLAP operations?

Business analysts perform several basic analytical operations with a multidimensional online analytical processing (MOLAP) cube.

Roll up

In roll up, the online analytical processing (OLAP) system summarizes the data for specific attributes. In other words, it shows less-detailed data. For example, you might view product sales according to New York, California, London, and Tokyo. A roll-up operation would provide a view of the sales data based on countries, such as the US, the UK, and Japan.

Drill down

Drill down is the opposite of the roll-up operation. Business analysts move downward in the concept hierarchy and extract the details they require. For example, they can move from viewing sales data by years to visualizing it by months.

Slice

Data engineers use the slice operation to create a two-dimensional view from the OLAP cube. For example, a MOLAP cube sorts data according to products, cities, and months. By slicing the cube, data engineers can create a spreadsheet-like table consisting of products and cities for a specific month.

Dice

Data engineers use the dice operation to create a smaller subcube from an OLAP cube. They determine the required dimensions and build a smaller cube from the original hypercube.

Pivot

The pivot operation involves rotating the OLAP cube along one of its dimensions to get a different perspective on the multidimensional data model. For example, a three-dimensional OLAP cube has the following dimensions on the respective axes:

- X-axis—product
- Y-axis—location
- Z-axis—time

Upon a pivot, the OLAP cube has the following configuration:

- X-axis—location
- Y-axis—time
- Z-axis—product

How does OLAP compare with other data analytics methods?

Data mining

Data mining is analytics technology that processes large volumes of historical data to find patterns and insights. Business analysts use data-mining tools to discover relationships within the data and make accurate predictions of future trends.

OLAP and data mining

Online analytical processing (OLAP) is a database analysis technology that involves querying, extracting, and studying summarized data. On the other hand, data mining involves looking deeply into unprocessed information. For example, marketers could use data-mining tools to analyze user behaviors from records of every website visit. They might then use OLAP software to inspect those behaviors from various angles, such as duration, device, country, language, and browser type.

OLTP

Online transaction processing (OLTP) is a data technology that stores information quickly and reliably in a database. Data engineers use OLTP tools to store transactional data, such as financial records, service subscriptions, and customer feedback, in a relational database. OLTP systems involve creating, updating, and deleting records in relational tables.

OLAP and OLTP

OLTP is great for handling and storing multiple streams of transactions in databases. However, it cannot perform complex queries from the database. Therefore, business analysts use an OLAP system to analyze multidimensional data. For example, data scientists connect an [OLTP database to a cloud-based OLAP cube](#) to perform compute-intensive queries on historical data.

How does AWS help with OLAP?

[AWS databases](#) provide various managed cloud databases to help organizations store and perform online analytical processing (OLAP) operations. Data analysts use AWS databases to build secure databases that align with their organization's requirements. Organizations migrate their business data to AWS databases because of the affordability and scalability.

- [Amazon Redshift](#) is a cloud data warehouse designed specifically for online analytical processing.
- [Amazon Relational Database Service \(Amazon RDS\)](#) is a relational database with OLAP functionality. Data engineers use Amazon RDS with Oracle OLAP to perform complex queries on dimensional cubes.
- [Amazon Aurora](#) is a MySQL- and PostgreSQL-compatible cloud relational database. It is optimized for running complex OLAP workloads.

21. DEMAND FOR ONLINE ANALYTICAL PROCESSING

In the top-down approach, you build the overall corporate-wide data repository using the entity-relationship (E-R) modeling technique. This enterprisewide data warehouse feeds the departmental data marts that are designed using the dimensional modeling technique. In the bottom-up approach, you build several data marts using the dimensional modeling technique and the collection of these data marts forms the data warehouse environment for your company. Each of these two approaches has its advantages and shortcomings. You also learned about a practical approach to building a conglomeration of supermarts with conformed and standardized data content. While adopting this approach, first you plan and define the requirements at the corporate level, build the infrastructure for the complete warehouse, and then implement one supermart at a time in a priority sequence. The supermarts are designed using the dimensional modeling technique. As we have seen, a data warehouse is meant for performing substantial analysis using the available data. The analysis leads to strategic decisions that are the major reasons for building data warehouses in the first place. For performing meaningful analysis, data must be cast in a way suitable for analysis of the values of key indicators over time along business dimensions. Data structures designed using the dimensional modeling technique support such analysis. In all the three approaches referred to above, the data marts rest on the dimensional model. Therefore, these data marts must be able to support dimensional analysis. In practice, these data marts seem to be adequate for basic analysis. However,

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in today's business conditions, we find that users need to go beyond such basic analysis. They must have the capability to perform far more complex analysis in less time. Let us examine how the traditional methods of analysis provided in a data warehouse are not sufficient and perceive what exactly is demanded by the users to stay competitive and to expand.

22. Limitations of Other Analysis Methods

You now have a fairly good grip on the types of requirements of users to execute queries and perform analysis. First and foremost, the information delivery system must be able to present multidimensional views of the data. Then the information delivery system must enable the users to use the data by analyzing it along multiple dimensions and their hierarchies in a myriad of ways. And this facility must be fast. It must be possible for the users to perform complex calculations. Let us understand why the traditional tools and methods are not up to the task when it comes to complex analysis and calculations. What information methods are we familiar with? Of course, the earliest method was the medium of reports. Then came spreadsheets with all their functionality and features. SQL has been the accepted interface for retrieving and manipulating data from relational databases. These methods are used in OLTP systems and in data warehouse environments. Now, when we discuss multidimensional analysis and complex calculations, how suitable are these traditional methods? First, let us look at the characteristics of the OLTP and data warehouse environments. When we mention the data warehouse environment here, we are not referring to heavy multidimensional analysis and complex calculations. We are only referring to the environment with simple queries and routine reports. Please see Figure 15-2 showing the characteristics of the OLTP and the basic data warehouse environments as they relate to information delivery needs. Now consider information retrieval and manipulation in these two environments. What are the standard methods of information delivery? Reports, spreadsheets, and online displays. What is the standard data access interface? SQL. Let us review these and determine if they are adequate for multidimensional analysis and complex calculations. Report writers provide two key functions: the ability to point and click for generating and issuing SQL calls, and the capability to format the output reports. However, report writers do not support

multidimensionality. With basic report writers, you cannot drill down to lower levels in the dimensions. That will have to come from additional reports. You cannot rotate the results by switching rows and columns. The report writers do not provide aggregate navigation. Once the report is formatted and run, you cannot alter the presentation of the result data sets. If report writers are not the tools or methods we are looking for, how about spreadsheets for calculations and the other features needed for analysis? Spreadsheets, when they first appeared, were positioned as analysis tools. You can perform “what if” analysis with spreadsheets.

23. OLAP Definitions and Rules

OLAP (Online Analytical Processing) is data processing technique. It is used to query and analyze data. It analyze data in real time from multiple sources.

In 1993, EF Codd introduced 12 rules for the relational database model for OLAP systems. These rules are known as "Codd's rules". These rules serve as a guideline for the design and development of OLAP systems. In this article, we will explore these 12 rules developed by Codd.

Codd's rules

These are 12 rules for OLAP systems, as proposed by E.F. Codd –

Codd's Rule

- 
- A diagram showing 12 rules of Codd's Rule. A vertical line on the left has 12 horizontal arrows pointing to the right, each pointing to a rule number and its name.
- 1. Information Representation
 - 2. Guaranteed Access
 - 3. Systematic Treatment of Null Values
 - 4. Database Description Rule
 - 5. Comprehensive Data Sub-Language
 - 6. View Updating
 - 7. High-Level Update, Insert, Delete
 - 8. Physical Data Independence
 - 9. Logical Data Independence
 - 10. The Distribution Rule
 - 11. Non-Subversion
 - 12. Integrity Rule

Multidimensional Conceptual View

OLAP system should provide multidimensional conceptual view of the data. This view should be based on the needs of the user and not on the physical data storage.

Transparency

System should be transparent to the user. But data abstraction property should be preserved.

Accessibility

System should provide easy access to the data. Users can access the data through user-friendly interface.

Consistent Reporting Performance

System should provide consistent reporting performance regardless of the complexity of the query or the amount of data being analyzed.

Client-Server Architecture

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System should be based on client-server architecture. It has multiple users to access the system at the same time.

Generic Dimensionality

The system should support generic dimensionality. System can handle any number of dimensions and any type of data.

Dynamic Sparse Matrix Handling

System should be able to handle dynamic sparse matrices. System can handle data that is not regularly populated.

Multi-User Support

System should support multi-user access. Multiple users can access and analyze data at the same time.

Unrestricted Cross-Dimensional Operations

System should allow unrestricted cross-dimensional operations. System should allow users to analyze data from different dimensions without restrictions.

Intuitive Data Manipulation

System should provide intuitive data manipulation tools. Users can manipulate and analyze data in a user-friendly manner.

Flexible Reporting

System should provide flexible reporting capabilities. Users can generate reports in various formats and with different levels of detail.

Unlimited Dimensions and Aggregation Levels

System should support unlimited dimensions and aggregation levels. System should can handle any number of dimensions and any level of aggregation.

OLAP characteristics, major features and functions

Online Analytical Processing (OLAP) is a powerful technology that has revolutionized data analysis and decision-making in various industries. In this blog, we delve into the fundamental concepts of OLAP, its types, advantages, architecture, and real-world applications. Discover what is OLAP in data mining, and how OLAP enables organizations to explore multidimensional data, uncover hidden insights, and drive informed strategies.

What is OLAP in a Data Warehouse?

OLAP, which stands for Online Analytical Processing, is a technology used in data analysis and business intelligence. It allows users to interactively analyze large volumes of multidimensional data in real-time. OLAP systems provide a way to organize, retrieve, and analyze data from various dimensions or perspectives, enabling users to gain insights and make informed decisions.

The fundamental concept of OLAP in data mining revolves around using multidimensional data structures OLAP allows users to drill down, roll up, slice, and dice data to explore different levels of detail and perspectives. To learn more about OLAP consider taking an online

Characteristics of OLAP

OLAP possesses several key characteristics that make it a valuable technology for data analysis and business intelligence. Here are the main characteristics of OLAP:

1. **Multidimensional Data Representation:** OLAP systems organize data in a multidimensional format, where different dimensions (such as time, geography, products, and more) are interconnected. This allows users to analyze data from various perspectives, enabling deeper insights.
2. **Cubes and Hypercubes:** OLAP systems use structures called “cubes” or “hypercubes” to store and represent data. These cubes contain aggregated and pre-calculated values for different combinations of dimensions, facilitating quick query response times.
3. **Dimension Hierarchies:** Dimensions within OLAP cubes often have hierarchies. For instance, a time dimension might have hierarchies like year > quarter > month > day. These hierarchies allow users to navigate and analyze data at different levels of granularity.
4. **Fast Query Performance:** OLAP systems are optimized for quick query performance. Aggregated data stored in cubes, along with indexing and pre-calculations, enables rapid response times for analytical queries, even over large datasets.
5. **Drill-down and Roll-Up:** Users can “drill down” to view more detailed data or “roll up” to see higher-level summaries. This capability to navigate between different levels of granularity aids in exploring data relationships.
6. **Slicing and Dicing:** “Slicing” involves selecting a specific value along one dimension to view a cross-section of data. “Dicing” involves selecting specific values along multiple dimensions. These operations allow users to focus on specific subsets of data.
7. **Advanced Calculations:** OLAP systems support various calculations beyond simple aggregation, such as ratios, percentages, and moving averages. These calculations aid in deriving meaningful insights from the data.
8. **User-Friendly Interface:** OLAP systems typically come with user-friendly interfaces that facilitate intuitive navigation and exploration of data. This makes it easier for non-technical users to perform complex analyses.
9. **Business Intelligence Integration:** OLAP is often integrated with business intelligence (BI) tools and reporting platforms. This integration allows users to create interactive dashboards, reports, and visualizations based on OLAP data.

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10. **Ad-Hoc Queries:** Users can create ad-hoc queries to answer specific analytical questions without needing to follow a predetermined query path. This flexibility is crucial for exploring unexpected insights.

Advantages of OLAP

OLAP offers several advantages that make it a valuable technology for data analysis and business intelligence. Here are the main advantages of OLAP:

1. **Interactive Analysis:** OLAP provides an interactive environment for users to explore data from various dimensions and perspectives. Users can drill down, roll up, slice, dice, and pivot data to gain insights and answer specific analytical questions.
2. **Quick Decision-Making:** With fast query response times and interactive capabilities, OLAP empowers users to make quicker and more informed decisions. This is especially important in fast-paced business environments.
3. **Flexibility:** OLAP allows users to create ad-hoc queries and modify queries on the fly to address new questions as they arise. This flexibility is crucial for exploring unexpected trends and patterns.
4. **Reduced Data Complexity:** OLAP systems abstract the complexity of underlying data structures by providing a user-friendly interface. Users can focus on exploring insights rather than dealing with complex database queries.
5. **Enhanced Collaboration:** OLAP's interactive nature facilitates collaboration among team members. Different users can explore the same data from their own perspectives and contribute insights to discussions.

Types of OLAP

There are various varieties of OLAP, each serving particular requirements and preferences for data analysis. The primary OLAP kinds are:

1. **MOLAP (Multidimensional OLAP):** MOLAP (Multidimensional OLAP) systems store data in a multidimensional cube structure, with aggregated data based on several dimensions contained in each cell of the cube. MOLAP systems do precalculations and store aggregations, which results in quick query responses. They work effectively in situations when performance is crucial and data quantities aren't very huge. Microsoft Analysis Services, IBM Cognos TM1, and Essbase are a few MOLAP system examples.
2. **Relational OLAP (ROLAP):** Traditional relational databases are used for data storage by ROLAP systems. They run intricate SQL queries to simulate multidimensional views of the data. ROLAP systems can manage huge datasets and complicated data linkages, therefore they can have slightly slower query speed than MOLAP systems, but they also provide better flexibility and scalability. ROLAP systems include those from Oracle OLAP, SAP BW (Business Warehouse), and Pentaho, as examples.
3. **Hybrid OLAP (HOLAP):** HOLAP systems attempt to combine the benefits of MOLAP and ROLAP. Similar to MOLAP, they enable the ability to obtain detailed data from the underlying relational database as necessary while also storing summary data in cubes. Depending on the type of analysis, this method helps to improve both performance and flexibility. Users of some MOLAP systems have the option of retrieving detailed data or pre-aggregated data by using HOLAP capabilities that are supported by these systems.
4. **DOLAP (Desktop OLAP):** Desktop OLAP, often known as DOLAP, is a simplified form of OLAP that operates on individual desktop PCs. It is appropriate for lone analysts who wish to carry out fundamental data exploration and analysis without requiring a large IT infrastructure. In-memory processing is frequently used by DOLAP tools to deliver comparatively quick performance on tiny datasets. The PivotTable feature in Excel is an illustration of a DOLAP tool.

5. **WOLAP (Web OLAP):** WOLAP systems bring OLAP capabilities to web browsers, allowing users to access and analyze data through a web-based interface. This enables remote access, collaboration, and sharing of analytical insights. WOLAP systems often use a combination of MOLAP, ROLAP, or HOLAP architectures on the backend. Web-based BI tools like Tableau, Power BI, and Looker provide WOLAP features.

Hypercube

- Hypercube model designing is basically a top-down approach/process. So multidimensional databases data can be represented for an application using two types of cubes i.e. hypercube and multi-cube. As shown in the figure (HYPERCUBE) below all the data appears logically as a single cube in a hypercube. All the parts of the manifold have identical dimensionality which are been represented by this hypercube.

In hypercube each dimension belongs to only one cube. A dimension is normally owned by the hypercube. Hence, this simplicity makes it easier for the users to use and understand it. As a hypercube is an top-down process, the designing of hypercube includes three major steps. The three major steps included are as follows:

1. You need to first decide which process of the business you want to capture in these model, such as sales activity.
2. Now, you identify the values which you want to be captured, such as sales amount. This information is always numeric in nature.
3. Now, identify the granularity of the data i.e. the lowest level of the data which you want to capture, this elements are dimensions. Some of the common dimensions are geography, time, customer and product.

For example, in a retail data warehouse, a hypercube might have dimensions for product, time, and location, and a measure for sales. This would allow users to easily analyze sales data by product, time, and location, and to create reports and visualizations that show sales trends and patterns.

Hypercubes can be created using a technique called “OLAP” (Online Analytical Processing) which is a powerful tool for data mining and analysis. OLAP allows users to perform complex analytical queries on a data warehouse, such as drilling down, rolling up, and slicing and dicing data. Hypercubes are also used in data visualization tools to create interactive dashboards, charts and graphs that allow users to easily understand the data. Hypercubes can handle large amounts of data and can be used for reporting, forecasting and other data analysis tasks.

Whereas, in multi-cube model the data is been segmented into a set of smaller cubes. Each one of these is composed of a subset of an available dimensions, refer figure (multi-cube) to understand it better. These are basically used to handle multiple fact tables, each with different dimensionality. Like the hypercube model the dimensions

are not always owned by any one cube, instead they are available to all cubes or there can be some dimensions which does not belong to any cube. The dimensions can be part of the multi-cubes. Hence, making it more versatile and efficient. It's also an efficient way of storing the sparse data, also it reduces the pre-calculation database explosion effect. One of the drawback of the multi-cube is that it is less straight forward than the hypercube and can carry steeper learning curves. There are many systems which uses combination of both this approach i.e. hypercube and multi-cube. This is done by separating storage, processing and presentation layers. So it stores the data as multi-cube but presents it as hypercube.

Properties

- **Multi-dimensional:** Hypercubes are multi-dimensional data structures that allow for the analysis of data across multiple dimensions, such as time, geography, and product.
- **Hierarchical:** Hypercubes are hierarchical in nature, with each dimension being divided into multiple levels of granularity, allowing for drill-down analysis.
- **Sparse:** Hypercubes are sparse data structures, meaning that they only contain data for cells that have a non-zero value. This can help to reduce storage requirements.
- **Pre-calculated:** Hypercubes are pre-calculated, meaning that the data is already aggregated and summarized, making it easier and faster to retrieve and analyze.
- **OLAP compatible:** Hypercubes are typically created using OLAP, which is a powerful tool for data mining and analysis. This allows for complex analytical queries to be performed on a data warehouse.
- **Data visualization:** Hypercubes can be used to create interactive dashboards, charts, and graphs that make it easy for users to understand the data.
- **Dynamic:** Hypercubes can be dynamic in nature, allowing users to change the data dimensions and levels of granularity on the fly.
- **Scalable:** Hypercubes can handle large amounts of data, and can be used for reporting, forecasting, and other data analysis tasks.
- **Efficient:** Hypercubes are efficient in terms of storage and retrieval of data, as they only store non-zero values.
- **Flexible:** Hypercubes can be used for a variety of data analysis tasks, including reporting, forecasting, and data mining.

Advantages

- **Ease of use:** Hypercubes are simple and straightforward to use, making them easy for users to understand and navigate.
- **Multi-dimensional analysis:** Hypercubes allow for multi-dimensional analysis of data, which can provide more in-depth insights and understanding of trends and patterns.

- **OLAP compatibility:** Hypercubes can be created using OLAP, which is a powerful tool for data mining and analysis. This allows for complex analytical queries to be performed on a data warehouse.
- **Data visualization:** Hypercubes can be used to create interactive dashboards, charts, and graphs that make it easy for users to understand the data.
- **Handling large amount of data:** Hypercubes can handle large amounts of data and can be used for reporting, forecasting, and other data analysis tasks.

Disadvantages

- **Complexity:** Hypercubes can be complex to set up and maintain, especially for large data sets.
- **Limited scalability:** Hypercubes can become less efficient as the amount of data grows, which can make them less scalable.
- **Performance issues:** Hypercubes can experience performance issues when trying to access and analyze large amounts of data.
- **Limited Flexibility:** Hypercubes are designed for a specific data structure and may not be able to handle changes in the data structure or accommodate new data sources.
- **Costly:** Hypercubes can be costly to implement and maintain, especially for large organizations with complex data needs.

26. OLAP Operations: Drill-down and roll-up.

[OLAP](#) is a widely spread technology belonging to [Business Intelligence processes](#) developed to coordinate and analyze vast amounts of data. OLAP databases are stored in the form of multidimensional cubes where each cube comprises the data supposed relevant by a cube administrator. Through certain OLAP operations, a user is able to obtain a specified view of the cube and extract requisite information from it. So this way it's possible to get a necessary Pivot Table and Pivot Chart report.

General OLAP operations involve Drill-up, Drill-down, Pivot, and Slice-and-Dice. Here we'd like to expand the list and look through all possible OLAP operations with examples for data mining including slicing and dicing in OLAP.

But before defining what is OLAP operation, let's figure out what language is used in this process.

OLAP language

OLAP operations could be based on two OLAP languages: SQL and MDX.

SQL or *Structured Query Language* is a computer language developed to work in two dimensions in order to manage relational database and manipulate data.

MDX or *Multidimensional expressions* is a language for analytical queries expression. Its principle difference from SQL language is that MDX is able to reference multiple dimensions. Microsoft primarily invented MDX as a SQL extension.

These two languages are different and have their own peculiarities. However, OLAP operations using SQL and MDX languages are pretty similar.

Our product Ranet OLAP uses MDX query language, that is why today we made an accent on MDX OLAP operations example.

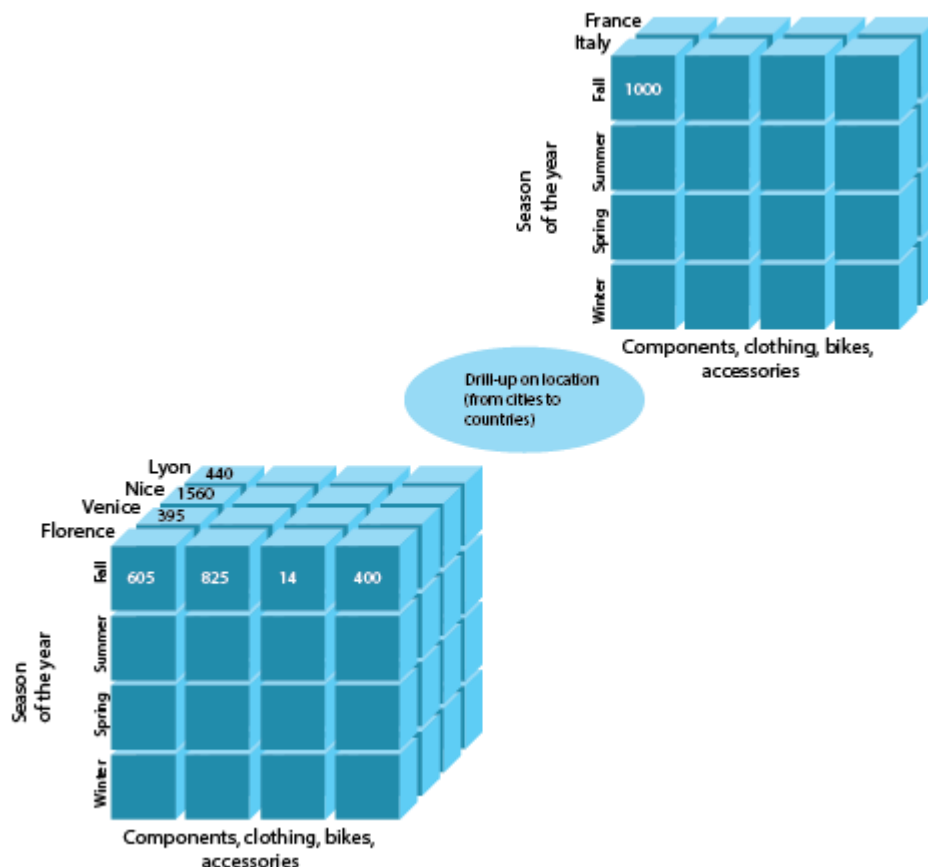
OLAP operations:

So let's outline the typical OLAP operations now.

Drill Up

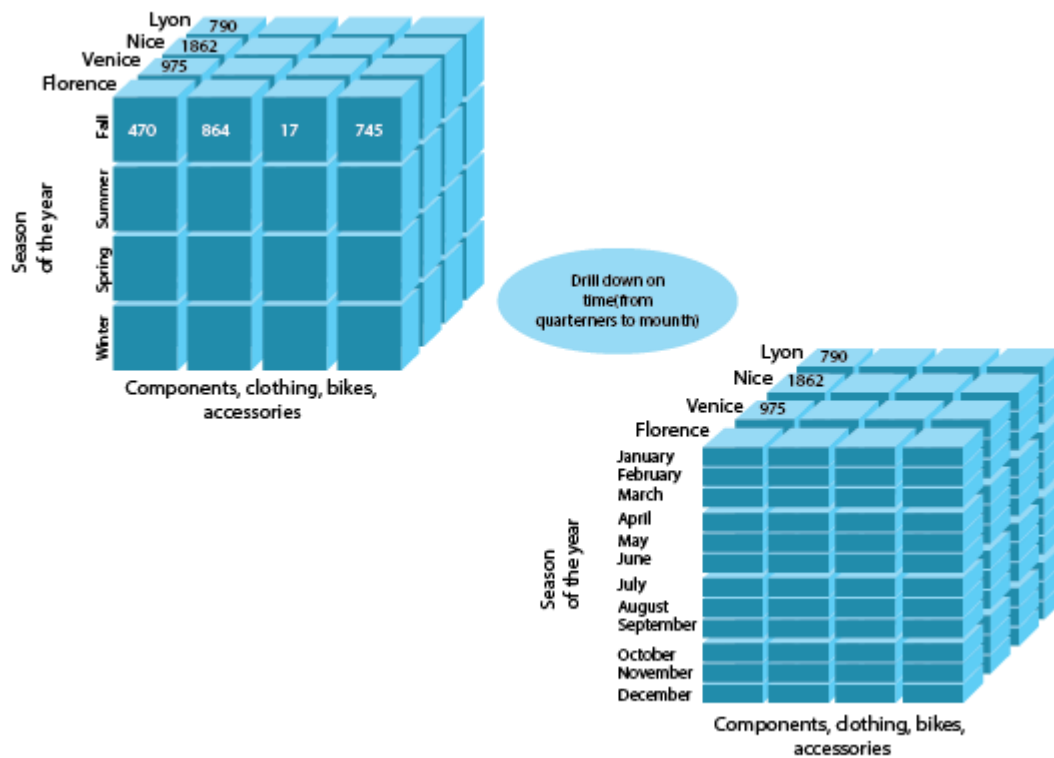
This operation you can meet as a part of pair drill up and drill down in OLAP. Drill-up is an operation to gather data from the cube either by ascending a concept hierarchy for a dimension or by dimension reduction in order to receive measures at a less detailed granularity. So that to see a broader perspective in compliance with the concept hierarchy a user has to group columns and unite the values. As there are fewer specifics, one or more dimensions from the data cube will be deleted, when this OLAP operation is run. In some sources drill up and roll up operations in OLAP come as synonyms, so this variant is also possible.

Here's a typical example of a Drill-up or roll up OLAP operations example:



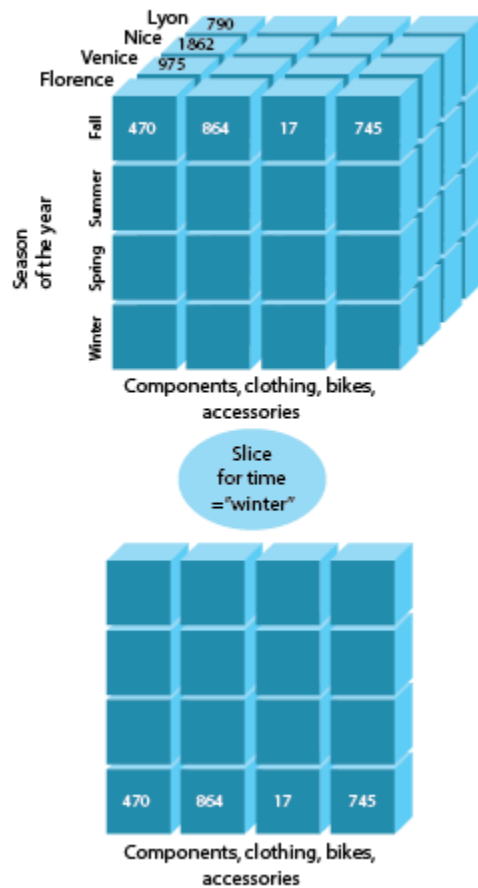
Drill down

OLAP Drill-down is an operation opposite to Drill-up. It is carried out either by descending a concept hierarchy for a dimension or by adding a new dimension. It lets a user deploy highly detailed data from a less detailed cube. Consequently, when the operation is run, one or more dimensions from the data cube must be appended to provide more information elements. Have a look at an OLAP Drill-down example in use:



Slice

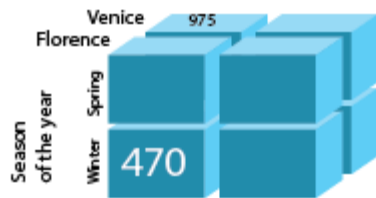
The next pair we are going to discuss is slice and dice operations in OLAP. The Slice OLAP operations takes one specific dimension from a cube given and represents a new sub-cube, which provides information from another point of view. It can create a new sub-cube by choosing one or more dimensions. The use of Slice implies the specified granularity level of the dimension. OLAP Slice example will look the following way:



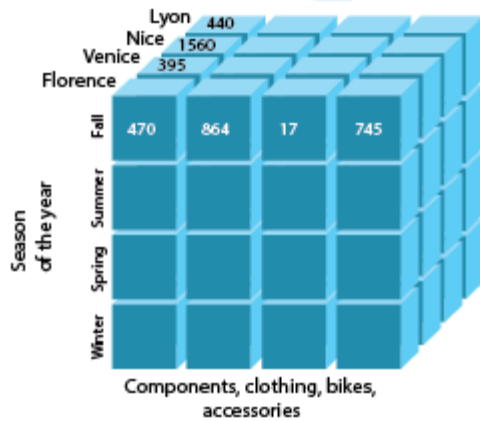
Dice

OLAP Dice emphasizes two or more dimensions from a cube given and suggests a new sub-cube, as well as Slice operation does. In order to locate a single value for a cube, it includes adding values for each dimension.

The diagram below shows how Dice operation works:



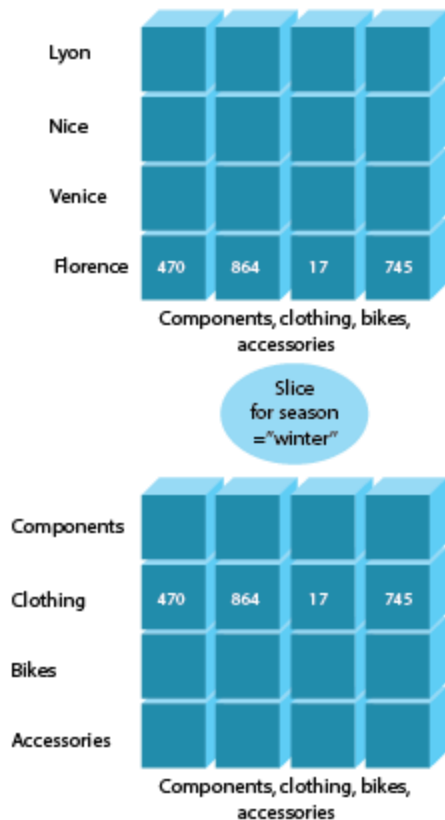
Dice for (location = "Venice" or "Florence") and (season = "Winter" or "Spring") and (item = "components" or "clothing")



Pivot

This OLAP operation rotates the axes of a cube to provide an alternative view of the data cube. Pivot clusters the data with other dimensions which helps analyze the performance of a company or enterprise.

Here's an example of Pivot in operation:



Scoping

The operation of Scoping restrains the presentation of the database objects to a specified subset. It will let users receive and update certain data values which they want. If there is a huge amount of data and a user needs to constrain the access of information to a specified subset Scoping is mostly conducive.

Screening

Screening is conducted to limit the set of data extracted.

Drill across

Drill across and Drill through in OLAP are another pair of opposite operations. The operation Drill across reconciles cells from several data cubes which share the same scheme.

Drill through

OLAP Drill through enables to navigate from data at the lower level in a cube to data in the operational systems whence the cube was ejected. The operation is usually exploited to identify the cause of outlier values in a data cube.

Sort

Sort brings the cube back where the members of a dimension were sorted.

Add Measure

Thanks to this OLAP operation one is able to add new measures to a cube.

Drop Measure

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In contrast to Add Measure, it's also possible to get rid of a measure from a data cube if it's not necessary.

Union

Due to an opportunity of Union, you can unite a number of cubes which have the same scheme but separate instances.

Difference

Difference eliminates the cells in a cube which are owned by another one. These two cubes must possess the same scheme.

ROLAP, MOLAP and HOLAP

•

1. Relational Online Analytical Processing (ROLAP) :

ROLAP servers are placed between relational backend server and client front-end tools. It uses relational or extended DBMS to store and manage warehouse data. ROLAP has basically 3 main components: Database Server, ROLAP server, and Front-end tool.

Advantages of ROLAP –

- ROLAP is used for handle the large amount of data.
- ROLAP tools don't use pre-calculated data cubes.
- Data can be stored efficiently.
- ROLAP can leverage functionalities inherent in the relational database.

Disadvantages of ROLAP –

- Performance of ROLAP can be slow.
- In ROALP, difficult to maintain aggregate tables.
- Limited by SQL functionalities.

2. Multidimensional Online Analytical Processing (MOLAP) :

MOLAP does not uses relational database to storage. It stores in optimized multidimensional array storage. The storage utilization may be low With multidimensional data stores. Many MOLAP server handle dense and sparse data sets by using two levels of data storage representation. MOLAP has 3 components : Database Server, MOLAP server, and Front-end tool.

Advantages of MOLAP –

- MOLAP is basically used for complex calculations.
- MOLAP is optimal for operation such as slice and dice.
- MOLAP allows fastest indexing to the pre-computed summarized data.

Disadvantages of MOLAP –

- MOLAP can't handle large amount of data.
- In MOLAP, Requires additional investment.
- Without re-aggregation, difficult to change dimension.

3. Hybrid Online Analytical Processing (HOLAP) :

Hybrid is a combination of both ROLAP and MOLAP. It offers functionalities of both ROLAP and as well as MOLAP like faster computation of MOLAP and higher scalability of ROLAP. The aggregations are stored separately in MOLAP store. Its server allows storing the large data volumes of detailed information.

Advantages of HOLAP –

- HOLAP provides the functionalities of both MOLAP and ROLAP.
- HOLAP provides fast access at all levels of aggregation.

Disadvantages of HOLAP –

HOLAP architecture is very complex to understand because it supports both MOLAP and ROLAP.

Difference between ROLAP, MOLAP and HOLAP :

Basis	ROLAP	MOLAP	HOLAP
Storage location for summary aggregation	Relational Database is used as storage location for summary aggregation.	Multidimensional Database is used as storage location for summary aggregation.	Multidimensional Database is used as storage location for summary aggregation.
Processing time	Processing time of ROLAP is very slow.	Processing time of MOLAP is fast.	Processing time of HOLAP is fast.
Storage space requirement	Large storage space requirement in ROLAP as compare to MOLAP and HOLAP.	Medium storage space requirement in MOLAP as compare to ROLAP and HOLAP.	Small storage space requirement in HOLAP as compare to MOLAP and ROLAP.
Storage location for detail data	Relational database is used as storage location for detail data.	Multidimensional database is used as storage location for detail data.	Relational database is used as storage location for detail data.

Basis	ROLAP	MOLAP	HOLAP
Latency	Low latency in ROLAP as compare to MOLAP and HOLAP.	High latency in MOLAP as compare to ROLAP and HOLAP.	Medium latency in HOLAP as compare to MOLAP and ROLAP.
Query response time	Slow query response time in ROLAP as compare to MOLAP and HOLAP.	Fast query response time in MOLAP as compare to ROLAP and HOLAP.	Medium query response time in HOLAP as compare to MOLAP and ROLAP.

29 Implementation Preparation

OLAP benefits without any doubts seem to be very attractive. However, there are also some issues you can face using OLAP system. Consider these challenges before implementation:

1. *Lack of standardization*: every OLAP vendor has its own client interface;
2. *Scalability*: OLAP system is good for processing summary data, but not for large amounts of detailed data.

However, for some companies those issues have the back positive side and are not so critical in general.

Preparation steps before OLAP implementation are no less (if even more) important than the implementation itself. You work for OLAP and then OLAP works for you. Here are different OLAP implementation considerations:

Vision and strategy development	To choose and implement the most suitable solution your team need to define business objectives first the most important steps as only the clear understanding of what you need and how to get it can lead The next stage is also to identify the strategy. To be precise OLAP strategy should correspond an en strategy and goals. Discuss those issues with your team and develop a unified concept.
Data preparation	First of all, it is important to learn as much as possible about the system. That is why before preparing be transferred into the new system, you should check OLAP data characteristics first: • OLAP data is summarized; • OLAP data is more departmentalized comparing with data warehouse which serves corporate-wide • The system stores and uses less data than a data warehouse. Make sure that conditions suit you and start your data preparation then.
Vendor and platform choice	Summarizing all the information and your requirements, it's time to choose vendor and an OLAP sy The choice should be made regarding the kind of the system: • ROLAP • MOLAP • HOLAP • Other

And of course, the vendor. As a little tip in this difficult choice we could offer a check list of what a system should have:

- ¾ Multidimensional data representation;
- ¾ Summarization, aggregation, derivations and precalculation capabilities;
- ¾ Formulas and complex calculations in an extensive library;
- ¾ Time intelligence;
- ¾ Cross-dimensional calculations;
- ¾ Simple interface.

Review

Summarize all the requirements, data and steps before implementation.

Implementation steps

To the time implementation of OLAP applications is about to start, you already know its features and functions, significance, and considerations. The only thing to do is summarize it all up. And, if you followed all the steps we mentioned before and spent your time for an appropriate preparation – congratulations, the major work is done. Now you can start implementation process easily. Just follow our final guidelines for OLAP implementation:

Step one: dimensional modeling

Step two: select the data required for removing into OLAP system

Step three: data extraction for the OLAP system

Step four: loading data to the OLAP server

Step five: data aggregation and derived data computation

Step six: implementation of OLAP application on desktop

Step seven: user's training organization

But how does it look in practice? Follow the next part and learn how OLAP implementation can be realized.

OLAP queries and reports

Characteristics of OLAP queries:

- Process huge data amounts;
- Support decision-making processes;
- Discover patterns and trends in the data.

OLAP language: OLAP query



The work with

OLAP is based on query. In order to operate report and analysis you have to post query, so it is rather relevant to talk about OLAP query language. The basic OLAP query language nowadays is MDX. *Multidimensional Expressions* (MDX) is a query language focused on access to multidimensional data structures. MDX queries are aimed at work with OLAP members, dimensions and hierarchies. With MDX it is possible to query data from SQL server and get a dataset with axis and cell data. Talking about MDX query language structure, it includes the following components:

- **SELECT:** the principle MDX statement. SELECT points out the dimension members that will be parts of axis;
- **WHERE:** identifies the dimension or member that are used as slicer dimensions;
- **WITH:** calculates the sets named during SELECT and WHERE clauses processing;
- **FROM:** determines the source of multidimensional data to which our query is addressed to.

The relevant question now is how to write OLAP queries? First of all, there are two types of them:

Simple

Contains only two MDX clause and answers a simple question. **Example:**

Complex

Can contain all kind of MDX clauses and answers complex

1 **SELECT**

```

1  SELECT
2  { ([Measures].[Unit Sales]) } ON
3  COLUMNS
FROM SALES

```

```

2  { [Measures].[Unit Sales], [Measures]
3  COLUMNS,
4  { [TIME].[1997], [TIME].[1998] } ON
5  FROM Sales
WHERE ( [Store].[USA].[CA] )

```

And if the simple MDX query can be posed by a user without technical background, to write complex query you need an additional IT help. For that reason, Ranet OLAP provides Query builder. This tool has two basic modes:

Custom query mode

This OLAP query manager suits perfectly for non-technical users. MDX query is generated by the system automatically and IT support is not required.

Design mode

Convenient for developers. The user creates query by drag and drop. The benefit is you can post a query of any complexity. The result will be based on your preferences.

OLAP Reports

OLAP and reporting

But what about OLAP reporting? Online analytical processing is about ad-hoc reporting primarily. In this way, ad-hoc report is a synonym for OLAP report. Let's highlight it in more detail.

OLAP reporting capacity

Reporting is one of the main purposes of OLAP. That is why the technology is all about to make this operation simple, understandable and available at any time. The remarkable characteristic of OLAP reports is that they are enable business users to get a response to any question on their demand. Those responses are based on the database connected to the application. This provides flexibility in constant changing business world as you are always have access to the relevant info any time you need it.

Reporting Tools

However, OLAP reporting requires knowledge of MDX or SQL languages, modern OLAP tools enable specialists without technical knowledge to use it too. What are OLAP reporting tools do exist? There are the following variants:

- **Simple pivot tables**

Pivot table designer is not used

The screenshot shows a reporting tool interface. On the left, there are sections for 'Filter' (with 'Calendar Year' selected), 'Rows' (with 'Country' selected), 'Columns' (with 'Values' selected), and 'Data' (with 'Internet Sales Amount', 'Internet Tax Amount', and 'Internet Order Count' selected). The main area displays a pivot table with 'Customer Geography' as the row label and the three measures as columns. The table is filtered by 'Country'. A 'Data Source Information' dialog box is open in the foreground, showing a custom MDX query. The query is as follows:

```

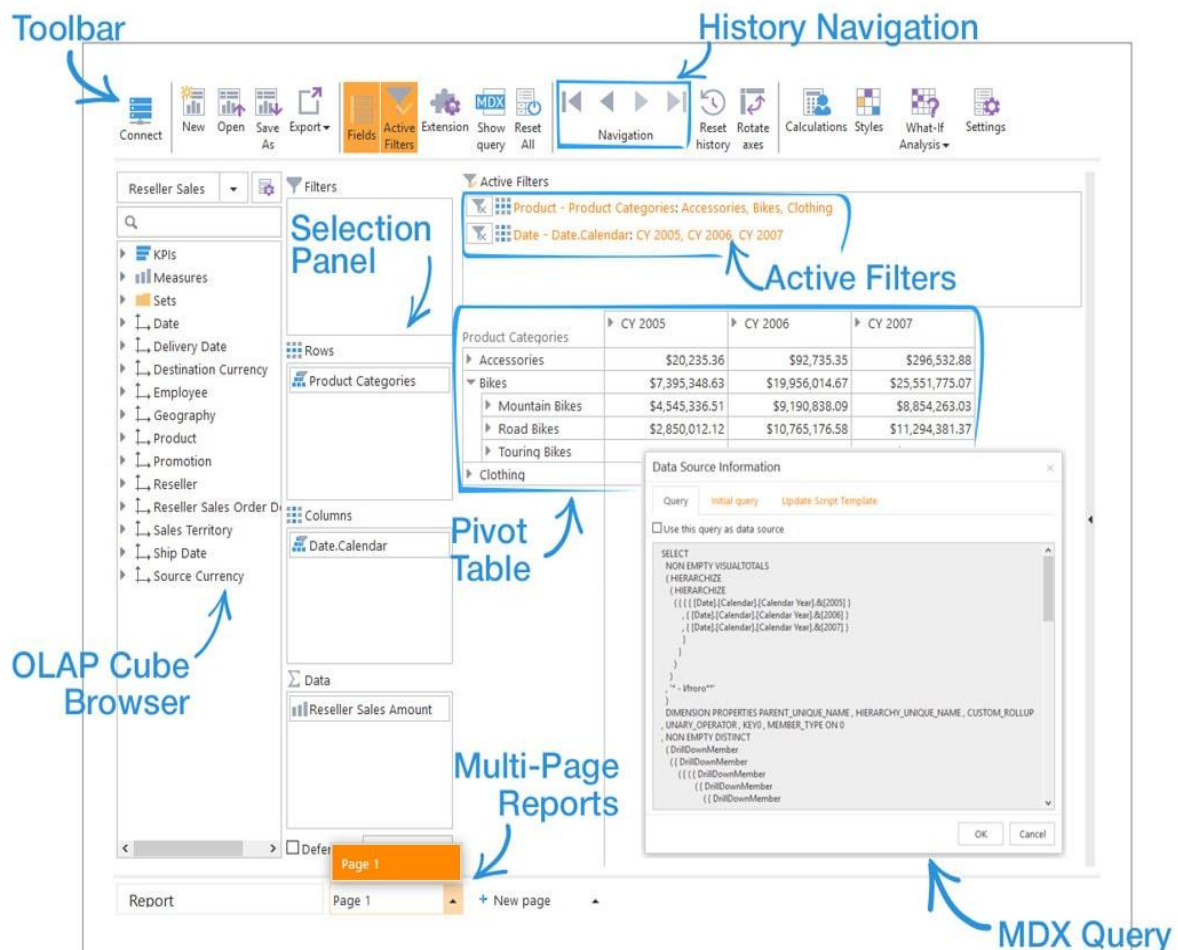
SELECT
  NON EMPTY VISUALTOTALS
  (
    HIERARCHIZE
    (
      {([Measures].[Internet Sales Amount],
        [Measures].[Internet Tax Amount],
        [Measures].[Internet Order Count])}
    ),
    "- Where*"
  )
  DIMENSION PROPERTIES PARENT_UNIQUE_NAME, HIERARCHY_UNIQUE_NAME, CUSTOM_ROLLUP
  , UNARY_OPERATOR, KEY0, MEMBER_TYPE ON 0
  , NON EMPTY DISTINCT
  (
    DrillDownMember
    (
      HIERARCHIZE
      (
        {([Customer].[Customer Geography].[Country].Members}
      )
    )
  )
)
  
```

At the bottom of the dialog, there are 'OK' and 'Cancel' buttons. A blue arrow points from the text 'Custom MDQ query initializes the pivot table' to the 'OK' button.

Custom MDQ query initializes the pivot table

Those reporting tools require the need to write MDX or SQL queries. But the plus is you can write queries of any complexity based on your needs.

- **Dynamic pivot tables**



You don't need any IT knowledge: just drag and drop elements into the special designer area and the system will do the rest itself. No need to write MDX query as well. Simple, interactive and easy-to use.

Moreover, no matter which tool you use, the reports can be simple one-page tables or complex interactive multi-page crosstab reports. They can have sorting, filtering, and drill through options, as well as various ways to visualize data on charts, heat maps.

Benefits

What can you get using OLAP reporting technology? There are following benefits you will enjoy:

- Simple and available: easy-to-use tools and possibility to make reports yourself makes it available for everyone;
- Data sharing: share your reports with colleagues and discuss it immediately;
- Flexible: reports creation in real time on users demand help to accommodate to new market conditions and client needs faster;
- Budget saving: no IT help is required as well as special trainings, so money on these recourses are saved.

OLAP reporting services example

All the above mentioned tools, benefits and features you can find in [Ranet OLAP](#) reporting tool. It provides capabilities for data analysis from different perspectives. Ranet OLAP includes visual

services, elements and controls for comprehensive business analysis. The product uses a single server platform and approach to UI implementation on HTML, WPF and Silverlight. Try it for free as Ranet suggests [30-day trial](#) and [demo](#). So check all the OLAP reporting possibilities yourself.

Important question relatef to OLAP report

1. What is an OLAP report?

OLAP usually uses the technology of ad-hoc reporting which is the model of real time reports generation used by the end-users on demand. OLAP reports ensure a response to a single specified question based on the database connected to the application. They are flexible and allows always get information right on time.

2. What is the distinguishing characteristic of OLAP reports?

The distinguishing characteristics of a good OLAP report are the following:

- They are flexible;
- Simple in usage;
- Save your time and money;
- Provide possibility: being generated by end-user or with SQL (or MDX) query written by IT specialist;

3. What are the top BI OLAP reporting tools?

There are a high number of OLAP vendors nowadays and major part of them correspond the characteristics mentioned above. However, Among OLAP vendors the following are the top among business users:

- IBM Cognos
- Flexmonster
- Oracle
- Sap
- Ranet OLAP

Executive Information System

What Does Executive Information System Mean?

An executive information system (EIS) is a decision support system (DSS) used to assist senior executives in the decision-making process. It does this by providing easy access to important data needed to achieve strategic goals in an organization. An EIS normally features graphical displays on an easy-to-use interface.

Advertisements

Executive information systems can be used in many different types of organizations to monitor enterprise performance as well as to identify opportunities and problems.

Techopedia Explains Executive Information System

Early executive information systems were developed as computer-based programs on mainframe computers to provide a company's description, sales performance and/or market research data for senior executives. However, senior executives were not all computer literate or confident. Moreover, EIS data was only supporting executive-level decisions but not necessarily supporting the entire company or enterprise.

Current EIS data is available company- or enterprise-wide, facilitated by personal computers and workstations on local area networks (LANs). Employees can access company data to help decision-making in their individual workplaces, departments, divisions, etc.. This allows employees to provide pertinent information and ideas both above and below their company level.

The typical EIS has four components: hardware, software, user interface and telecommunication.

What Are the Characteristics of an Ideal Executive Information System?

Not all company executives are tech-savvy. As such, an ideal EIS should have a graphical display that has an easy-to-use interface. Executives should be able to see all the variables and trends needed to make sound business decisions in a single dashboard. That way, they can make comparisons and projections to ensure growth.

Note, too, that each executive specializes in a particular business area. As such, an ideal EIS should have drill-down capabilities that will enable them to zoom in on areas that fall under their responsibility.

What Are the Components of an Executive Information System?

An EIS has four major components, which are:

- Hardware
- Software
- User interface (UI)
- Telecommunications capability

Hardware

An EIS's hardware should include input devices that executives can use to enter, check, and update data; a [central processing unit \(CPU\)](#) that controls the entire system; data storage for saving and archiving useful business information; and output devices (e.g., monitors, printers, etc.) that show visual representations of the data executives need to keep or read.

Software

An EIS's software should be able to integrate all available data into cohesive results. It should be capable of handling text and graphics; connected to a database that contains all relevant internal and external data; and have a model base that performs routine and special statistical, financial, and other quantitative analyses.

User interface (UI)

This component should be capable of producing scheduled reports, FAQs, and other information. It would be best if it's menu-driven, too, allowing executives to pick from predetermined choices for their needs. And since not all executives are tech-savvy, it's ideal for the UI to accept inputs and produce outputs using programming (i.e., for the tech-savvy) and natural language (i.e., for the not tech-savvy).

Telecommunications capability

Since most executives often travel, an EIS should have telecommunications capability. That way, it remains accessible regardless of location.

What Are the Benefits of Using an Executive Information System?

Using an EIS provides the following benefits:

- Easy for any executive to use
- Provides the ability to analyze trends
- Augments an executive's leadership capabilities
- Enhances personal thinking and decision-making
- Makes strategic control flexible
- Enhances an organization's market competitiveness
- Creates better reports
- Improves consensus building and communication
- Enables office automation
- Reduces time required to find information
- Enables company performance predictions
- Allows detailed examinations of critical success factors

What Are the Disadvantages of Using an Executive Information System?

Despite providing several advantages, EIS usage has cons, too, such as:

- Limited functionality
- Hard to quantify the benefits
- Possible information overload on an executive's part
- System may become slow over time

- May lead to system insecurities
- May be too expensive for small companies